

Research Article

Dual-Function Catalytic Regulation of Polysulfides by V₂CT_x/SnS₂ Heterostructures Enabling High-Performance Li–S Battery Separators

Submission ID a5878c84-dc2f-48bf-8d05-26c8947aa39a

Submission Version Initial Submission

PDF Generation 20 Aug 2026 07:12:35 EST by Atypon ReX

Authors

Mr. Longyu Liu

[CRediT](#)

Writing - original draft, Writing - review & editing, Data curation, Validation

Affiliations

- College of Photonics and Optical Engineering, Aerospace Information Technology University, Jinan, China
- School of Physics and Optoelectronic Engineering, Shandong University of Technology, Zibo, China

Prof. Fei Xing
Corresponding Author
Submitting Author

 [ORCID](#)

<https://orcid.org/0000-0001-8184-1343>

[CRediT](#)

Conceptualization, Resources, Writing - review & editing, Funding acquisition

Affiliations

- College of Photonics and Optical Engineering, Aerospace Information Technology University, Jinan, China
- School of Physics and Optoelectronic Engineering, Shandong University of Technology, Zibo, China

Mr. Songshi Xu

[CRediT](#)

Methodology, Investigation, Validation, Software

Affiliations

- School of Physics and Optoelectronic Engineering, Shandong University of Technology, Zibo, China

Ms. Yingfei Wen

[CRediT](#)

Validation, Writing - review & editing

Affiliations

- School of Physics and Optoelectronic Engineering, Shandong University of Technology, Zibo, China

Mr. Minghui Dong

CReditT

Formal analysis

Affiliations

- School of Materials Science and Engineering, Southeast University, Nanjing, China

Prof. Qi Fan

 ORCID

<https://orcid.org/0000-0001-9304-9417>

CRediT

Formal analysis, Supervision

Affiliations

- School of Materials Science and Engineering, Southeast University, Nanjing, China

Prof. Ze Zhang

Affiliations

- College of Photonics and Optical Engineering, Aerospace Information Technology University, Jinan, China
- Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China
- Key Laboratory of Computational Optical Imaging Technology, Chinese Academy of Sciences, Beijing, China
- Qilu Aerospace Information Research Institute, Jinan, China

CReditT

Formal analysis, Supervision

Files for peer review

All files submitted by the author for peer review are listed below. Files that could not be converted to PDF are indicated; reviewers are able to access them online.

Name	Type of File	Size	Page
Battery_Data_Reporting_Checklist-1643963769483.pdf	Data Reporting Checklist	1.3 MB	Page 4
Manuscript.docx	Main Document - MS Word	3.1 MB	Page 5
Supporting.docx	Supporting Information for Peer Review and Publication	4.6 MB	Page 33

WILEY-VCH

Battery Data Reporting Checklist

At Wiley-VCH, we wish to improve the reproducibility and data quality of our published content according to the [FAIR principles](#). Please fill out this form as part of the submission process for all original research articles reporting performance and/or stability-related investigations of energy storage devices (batteries). For further information please see Wiley-VCH’s [data sharing policy](#).

Components and Setup		
<i>Please provide the name of the material used and its composition (loading/amount if applicable)</i>		
Active material (please provide theoretical capacity)		
Additive		
Binder		
Electrolyte		
Current collector		
Separator		
<i>Please specify setup and components (check all that apply and/or select from the dropdown menu)</i>		
Setup	Full cell	Half cell
	Anode	Counter
	Cathode	Reference
Electrode thickness		Working
Comments		

Measurements and Calculations			
<i>Please show the calculation used to determine the capacity (reported in mAh/g)</i>			
Initial capacity (mAh/g)			
Capacity achieved (mAh/g)			
@rate	@cycle number		
Number of cycles tested			
Capacity retention			
Testing temperature			

Characterization (please select appropriate)			
Cyclic voltammetry		Electron microscopy	
Rate capability		X-ray characterization	
Charge/discharge tests		Electrochemical impedance spectroscopy	
Other			

This Battery Data Reporting Checklist is made available under the Creative Commons Attribution (CC-BY) License, which permits use, distribution and reproduction in any medium, provided the original checklist is properly cited. Please consult the related article for the license terms applicable to the article.

Dual-Function Catalytic Regulation of Polysulfides by V_2CT_x/SnS_2 Heterostructures Enabling High-Performance Li–S Battery Separators

Longyu Liu^{a,b,+}, Fei Xing^{a,b,*,+}, Songshi Xu^b, Yingfei Wen^b, Minghui Dong^c, Qi Fan^c, Ze Zhang^{a,d,e,f,*}

^aCollege of Photonics and Optical Engineering, Aerospace Information Technology University, Jinan, China.

^bSchool of Physics and Optoelectronic Engineering, Shandong University of Technology, Zibo, China.

^cSchool of Materials Science and Engineering, Southeast University, Nanjing, China.

^dAerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

^eKey Laboratory of Computational Optical Imaging Technology, Chinese Academy of Sciences, Beijing, China.

^fQilu Aerospace Information Research Institute, Jinan, China.

E-mail: Fei Xing(xingfei@sdu.edu.cn) and Ze Zhang(zhangze@aircas.ac.cn)

Keywords: lithium–sulfur batteries, V_2CT_x MXene, SnS_2 , heterostructures, polysulfide conversion, multifunctional separators

Abstract: Transition metal sulfides and two-dimensional materials have been a focal point in lithium-sulfur (Li-S) battery research. In this study, a novel bifunctional heterojunction catalyst composed of transition metal sulfides and two-dimensional materials was designed. This catalyst is a heterostructure formed by ellipsoidal SnS_2 and monolayer V_2CT_x nanosheets. Under the influence of the Mott-Schottky effect, due to the availability of electrons in the sulfur atoms of SnS_2 and the tendency of vanadium (V) atoms in MXenes to lose electrons, SnS_2 and V_2CT_x spontaneously form a built-in electric field (BIEF). In this BIEF, electrons flow from V_2CT_x to SnS_2 , facilitating easier binding of SnS_2 with polysulfides (LiPSs). Additionally, the BIEF enhances the rate of Li^+ transfer, significantly lowering the nucleation and decomposition barriers of Li_2S . The Li-S battery with SnS_2 - V_2C heterostructure separators achieves a first-cycle discharge capacity of 1256.1 mAh g⁻¹ at 0.2 C and maintains excellent long-term cycling stability with a capacity decay rate of only 0.019% after 3000 cycles at 2 C. This study presents a simple method to fabricate heterostructures and demonstrates their catalytic performance in Li-S battery separators.

Longyu Liu and Fei Xing contributed equally to this work.

1. Introduction

With the rapid development and widespread adoption of new energy vehicles and electronic devices, energy-storage systems with long cycle life and high energy density have become increasingly critical. Lithium–sulfur (Li–S) batteries have attracted considerable attention owing to their exceptionally high theoretical specific capacity ($\sim 1675 \text{ mAh g}^{-1}$).^[1–3] However, practical commercialization of Li–S batteries remains hindered by several key issues, including the intrinsically low electrical conductivity of sulfur cathodes, the shuttle effect of soluble lithium polysulfides (LiPSs), sluggish sulfur conversion kinetics, electrolyte consumption and short-circuit risks induced by dendrite growth on the lithium metal anode, and severe volume variation during cycling. In particular, polysulfide migration and accumulation lead to active-material loss, aggravated polarization, and shortened cycle life, preventing Li–S systems from fully realizing their theoretical advantages.^[4–6] Previous efforts have largely focused on modifying and optimizing sulfur cathodes. Nevertheless, the separator plays a pivotal role in governing the shuttle effect in Li–S batteries. Therefore, introducing innovative functional composite separators represents an effective strategy to suppress polysulfide shuttling.

Early research on lithium-sulfur (Li-S) battery separator modification primarily focused on enhancing the functionality of separators through inorganic and polymer materials. Inorganic materials such as oxides, carbon nanotubes (CNTs), and graphene have been widely applied for separator modification due to their excellent conductivity and strong adsorption capabilities. These materials help alleviate the shuttle effect and reduce the solubility of lithium polysulfides (LiPSs) through physical interception and chemical adsorption.^[7]

However, these modified materials often suffer from drawbacks such as insufficient conductivity, structural instability, and high costs, which hinder their long-term stable performance in practical applications. Additionally, polymer materials, such as PVDF and PAN, have been used as separators or composites to enhance the mechanical strength and wettability of the separators. While polymers offer good mechanical stability, their poor conductivity and inability to effectively adsorb LiPSs limit their effectiveness in mitigating the shuttle effect in Li-S batteries. Especially during charge-discharge cycles, polymer separators may undergo swelling or rupture, which affects the cycling stability of the battery. To address these issues, MXene materials, a new class of two-dimensional materials, have garnered significant attention in recent years for their potential in modifying separators in Li-S batteries.^[8]

MXene is a novel class of two-dimensional materials that is typically prepared by selectively etching the “A” layer from the layered MAX phase ($M_{n+1}AX_n$, $n=1-4$). Here, “M” represents transition metal elements (such as Ti, V, Zr, and Nb), “A” denotes IIIA or IVA group elements (such as Al, Ga, Si), and “X” refers to carbon or nitrogen. After etching, the “A” layer is removed, resulting in the formation of $M_{n+1}X_nT_x$ MXene, which is functionalized with surface termination groups (such as O, OH, F, Cl). The type of surface termination group is determined by the etching agent used and can be further modified through post-treatment. To date, more than 30 MXenes have been synthesized, and the MXene family continues to expand. In various energy-storage devices—such as lithium-ion batteries, sodium-ion batteries, Li-S batteries, zinc-ion batteries, and supercapacitors—MXenes have demonstrated substantial potential. Their properties can be tailored by adjusting composition, structure, and surface terminations. For high-performance Li-S batteries,

MXenes are particularly promising. As quasi-metallic conductors, MXenes provide rapid electron transport, improving sulfur utilization and enabling high specific capacity; thus, they exhibit intrinsic advantages over conventional 2D materials for addressing key challenges in Li–S systems.

To enhance the electrochemical performance of Li–S batteries, we propose a separator-integrated heterostructure composed of SnS₂ and V₂CT_x MXene. Compared with the widely studied Ti₃C₂T_x MXene, V₂CT_x is a less widely explored MXene that offers distinct advantages in Li–S batteries due to its unique chemical and structural features.^[9–11] First, surface V atoms on V₂C MXene interact more strongly with LiPSs, enabling more effective adsorption and thereby suppressing the shuttle effect and reducing active-material loss. Furthermore, V₂CT_x exhibits higher electrical conductivity, benefiting from its favorable electronic structure, which markedly improves electron transport at the sulfur cathode and enhances rate capability and capacity utilization. In addition, vanadium possesses outstanding catalytic activity, which can accelerate the conversion of LiPSs to Li₂S₂/Li₂S, reduce polarization, and increase reaction efficiency. Meanwhile, the layered architecture of V₂CT_x is more stable during cycling, accommodating the volume change of sulfur cathodes and extending cycle life. The results show that V₂CT_x MXene has a wider electrochemical window and richer surface chemical characteristics, enabling more efficient sulfur loading and the formation of a stable interface, thereby enhancing the battery's specific capacity and overall performance. These attributes collectively endow V₂CT_x MXene with strong application potential in Li–S batteries. However, V₂CT_x is a layered 2D material that tends to restack in practical use, leading to reduced surface area and fewer accessible active sites. To address this issue, we coupled SnS₂ with V₂C.^[12] SnS₂, featuring a hexagonal lattice and Sn–S

bonding, can act as a structural spacer to prevent V_2C restacking, while V_2C provides fast conductive pathways to compensate for the relatively poor conductivity of sulfides.^[13–17] This complementary integration prevents mutual active-site shielding and strengthens LiPS capture. Moreover, owing to the difference in work function, when SnS_2 contacts V_2CT_x , the Fermi-level mismatch drives electron transfer from V_2CT_x to SnS_2 , inducing interfacial band bending and forming a spontaneous BIEF. The rectifying interfacial field directs electron transport, promotes faster Li^+ migration toward the cathode, and—more importantly—enhances the adsorption and catalytic reactivity of SnS_2 , facilitating the solid–liquid conversion of LiPSs during cycling and improving reaction kinetics. Meanwhile, SnS_2 can effectively reduce electrochemical polarization and enhance the redox capacity of LiPSs.^[18,19] The sulfur affinity of SnS_2 provides an inherent advantage, which can accelerate the conversion of LiPSs and reduce their dissolution, thereby alleviating the shuttle effect.

2. Results and Discussion

2.1. Catalytic Kinetics of V_2CT_x/SnS_2 Heterostructures in Lithium–Sulfur Batteries

Figure 1 presents multiple microscopic characterizations of the V_2CT_x/SnS_2 composite.

Figures 1a–c show SEM images of V_2CT_x , SnS_2 , and the V_2CT_x/SnS_2 composite, respectively.

In Figure 1a, V_2CT_x exhibits a typical layered morphology with flat and smooth etched nanosheets, limited overlap between sheets, and uniform, well-defined flake sizes. In Figure 1b, SnS_2 shows a nanosphere-like morphology; however, evident aggregation is observed, which may reduce the exposure of active sites. Figure 1c displays the composite after coupling V_2CT_x with SnS_2 : SnS_2 nanoparticles are uniformly distributed on the V_2CT_x surface, and the V_2CT_x surface becomes rougher.^[20]

Figure 1d shows a TEM image revealing the overall architecture of the composite. A clear interface between V_2CT_x and SnS_2 confirms effective hybridization. Figure 1e and 1f present HRTEM images of enlarged regions, showing distinct lattice spacings. The (110) plane of V_2CT_x exhibits a lattice spacing of 0.35 nm, while the (001) plane of SnS_2 shows 0.625 nm. The lattice matching further evidences intimate interfacial contact and the formation of a heterojunction; this is consistent with the characterization results of similar studies.^[21]

Figure 1g shows HAADF-STEM and elemental mapping of the composite, demonstrating uniform distributions of C, V, Sn, and S, further confirming successful hybridization and a stable interfacial structure with abundant catalytic/adsorptive active sites.

The phase structure of V_2CT_x/SnS_2 , V_2AlC , V_2CT_x , and SnS_2 was analyzed by X-ray diffraction (XRD) (Figure S1).^[22,23] V_2CT_x exhibits distinct peaks at 13.3° (002) and 41.1° (104). After etching with HF and HCl, the Al peak corresponding to (104) disappears, and the (002) peak shifts to a lower angle at 7.2° and broadens, indicating that the etched V_2CT_x captures surface functional groups, leading to an expansion of the interlayer spacing and successful transformation of the MAX phase into V_2CT_x nanosheets. For the V_2CT_x/SnS_2 composite material, the (002) diffraction peak further shifts to the left, suggesting that SnS_2 successfully inserted between the V_2CT_x nanosheets, further expanding the interlayer spacing. SnS_2 exhibits two characteristic peaks at 14.9° and 50.0° , corresponding to its (001) and (110) crystal planes. Furthermore, the composite material shows the characteristic diffraction peaks of both V_2CT_x and SnS_2 , providing further evidence of the successful integration of the two materials .

Overall, coupling V_2CT_x with SnS_2 yields a function-rich heterojunction with strong catalytic and adsorption capabilities. V_2CT_x serves as a conductive scaffold to enhance electrical

conductivity, while SnS_2 functions as a catalyst to promote LiPS conversion. Their synergy can effectively suppress the shuttle effect, improve electrochemical performance, and enhance cycling stability and rate capability in Li–S batteries.

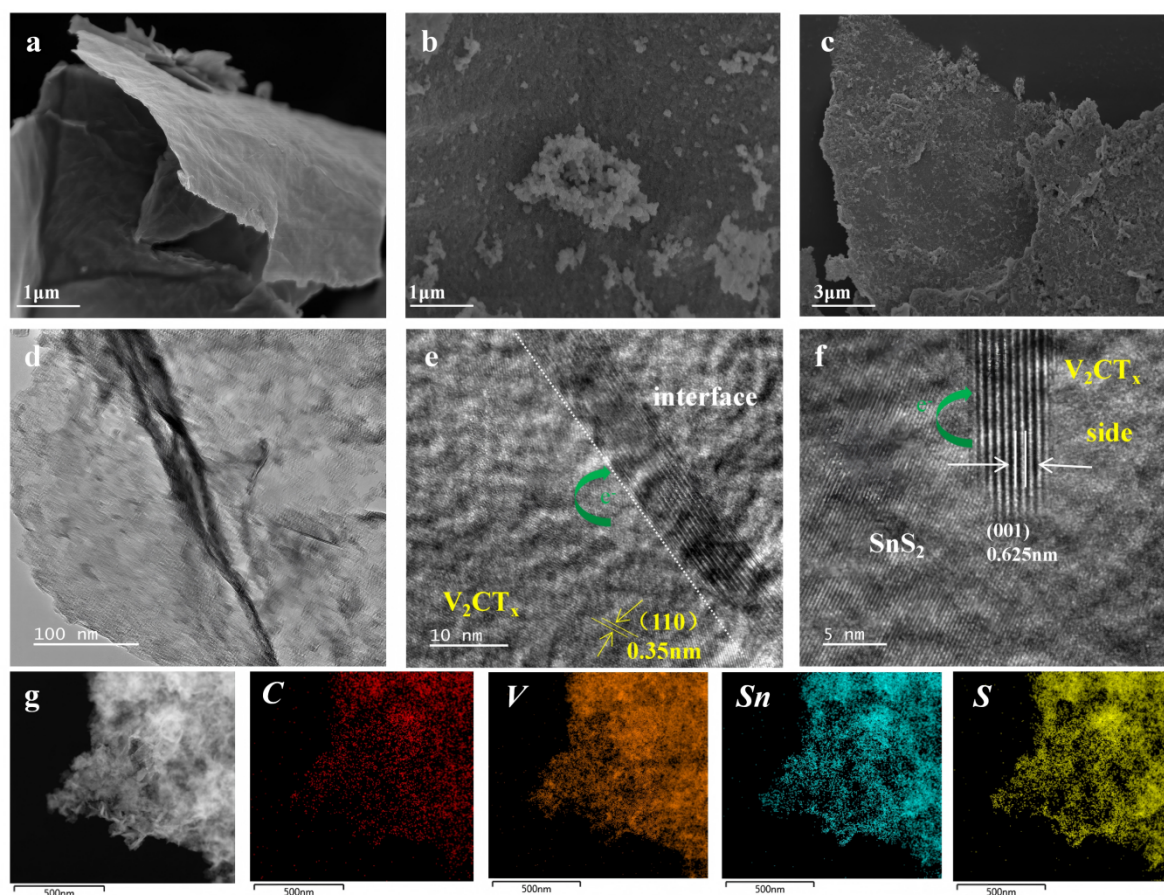


Figure 1. (a) SEM image of V_2CT_x . (b) SEM image of SnS_2 . (c) SEM image of $\text{V}_2\text{CT}_x/\text{SnS}_2$. (d) TEM image of $\text{V}_2\text{CT}_x/\text{SnS}_2$. (e, f) HRTEM images of $\text{V}_2\text{CT}_x/\text{SnS}_2$. (g) HAADF-STEM image and corresponding elemental maps of C, V, Sn and S.

XPS of V_2CT_x (**Figure 2a**) and the fitted V 2p–O 1s spectra show that V 2p can be deconvoluted into four doublets: peaks at 524.10 eV and 516.60 eV are assigned to V^{5+} ; 522.20 eV and 514.7 eV to V^{4+} ; 521.15 eV and 513.5 eV to V^{3+} ; and 520.20 eV and 512.7 eV to V^{2+} . The characteristic spectra indicate an increase in high-valence V, attributed to further oxidation during synthesis. High-valence V can bind LiPSs more strongly, stabilizing soluble LiPS species. The Sn 3d spectrum of $\text{V}_2\text{CT}_x/\text{SnS}_2$ (Figure S2) resembles that of SnS_2 , with two main peaks at 495.2 eV ($\text{Sn } 3d_{5/2}$) and 486.8 eV ($\text{Sn } 3d_{3/2}$). Compared with pristine SnS_2 ,

the Sn 3d peaks in V_2CT_x/SnS_2 shift to lower binding energies, while the S 2p_{1/2} peak broadens (Figure S3), indicating strong interfacial interaction between SnS_2 and V_2CT_x . The survey spectrum confirms the coexistence of elements from both components (Figure S4), consistent with elemental mapping.^[24–26]

Figure S5 demonstrates that, through electrostatic potential calculations, the work function values of V_2CT_x and SnS_2 are 4.06 eV and 6.34 eV, respectively. Upon contact between the two materials, the difference in work functions causes electron flow from V_2CT_x (metal) to SnS_2 (semiconductor) until their Fermi levels align (Figures 2b and 2c). During this process, the electron movement induces the formation of a BIEF at the interface, driving electrons from V_2CT_x to SnS_2 , which leads to band bending at the interface.^[27] As electrons accumulate on the SnS_2 surface, the semiconductor nature of SnS_2 facilitates electron enrichment, promoting chemical bonding with additional LiPSs, thereby effectively capturing more LiPSs. During the nucleation or decomposition of Li_2S , the abundant Li^+ and electrons are able to rapidly migrate across the heterojunction interface, facilitating fast Li^+/e^- transfer and significantly reducing the nucleation/decomposition barrier of Li_2S . This process ultimately accelerates the bidirectional sulfur conversion kinetics. Consequently, the interface between SnS_2 and V_2CT_x exhibits characteristics of an Ohmic contact, wherein electrons flow freely without encountering significant energy barriers.

The interaction between the material and LiPSs is a key factor influencing catalytic conversion. To investigate this, static adsorption experiments of Li_2S_6 and UV-Vis tests were conducted. As shown in the inset of Figure 2e, the Li_2S_6 solution containing V_2CT_x/SnS_2 material completely decolorizes within 6 hours, indicating the successful adsorption of Li_2S_6 , whereas the Li_2S_6 solution containing only V_2CT_x turns dark brown. This color change is due

to the dissolution of V_2CT_x , which appears black and mixes with the yellow of Li_2S_6 , demonstrating that Li_2S_6 was not fully adsorbed. In contrast, the solution containing V_2CT_x/SnS_2 turns transparent. To further verify this, UV-Vis testing was performed on the post-reaction solution. As clearly seen in Figure 2e, the V_2CT_x/SnS_2 material exhibits the lowest peak intensity, indicating that the material adsorbed Li_2S_6 , causing a decrease in its concentration. Moreover, the V_2CT_x/SnS_2 material shows the weakest absorption peak. These results indicate that the composite material has a strong adsorption capability toward LiPSs.^[28]

To investigate the adsorption interaction between SnS_2 and LiPSs, the crystal structure of SnS_2 was optimized, and the adsorption energies between LiPSs and the SnS_2 surface were calculated using density functional theory (DFT), based on the chemical characteristics of polysulfides (Figure S6). The results show that the adsorption energies of SnS_2 for Li_2S_8 , Li_2S_6 , and Li_2S_4 are -1.11 eV, -0.87 eV, and -0.82 eV, respectively. This suggests that tin sulfide has a strong chemical interaction with polysulfides, and this strong adsorption further inhibits the shuttle effect. This interaction helps anchor LiPSs onto the V_2CT_x/SnS_2 heterojunction and reduces their dissolution into the electrolyte. Further electronic structure analysis reveals that the interaction between the SnS_2 surface and LiPSs molecules primarily occurs through electron transfer and chemical bonding (Figure 2d and Figure S6). This electron transfer process contributes to the stability of polysulfides and may suppress their shuttle effect in the battery.^[29]

An H-type diffusion cell was used to study Li_2S_6 permeation through different separators. One side contained 4 mM Li_2S_6 and the other contained pure DOL/DME; the two chambers were separated by the corresponding separator. Results (Figure S7) show that pristine PP poorly

blocks LiPSs: the right chamber becomes pale yellow after 1 h, deepens after 2 h, and stabilizes after 8 h, indicating rapid diffusion through PP. PP@V₂CT_x partially suppresses permeation, with the right chamber color no longer deepening after 12 h. PP@V₂CT_x/SnS₂ exhibits the slowest color change, and even after 12 h the right chamber remains much lighter than in the other cases, demonstrating effective suppression of LiPS permeation. Thus, the blocking capability follows: PP@V₂CT_x/SnS₂ > PP@V₂CT_x > PP. Replacing PP with PP@V₂CT_x/SnS₂ provides combined physical/chemical protection, markedly mitigating shuttle and stabilizing electrochemical reactions. Moreover, the SnS₂@V₂C-loaded separator exhibits good mechanical robustness and adhesion stability, maintaining morphology after repeated bending (Figure S8), which helps buffer volume-change effects during cycling.

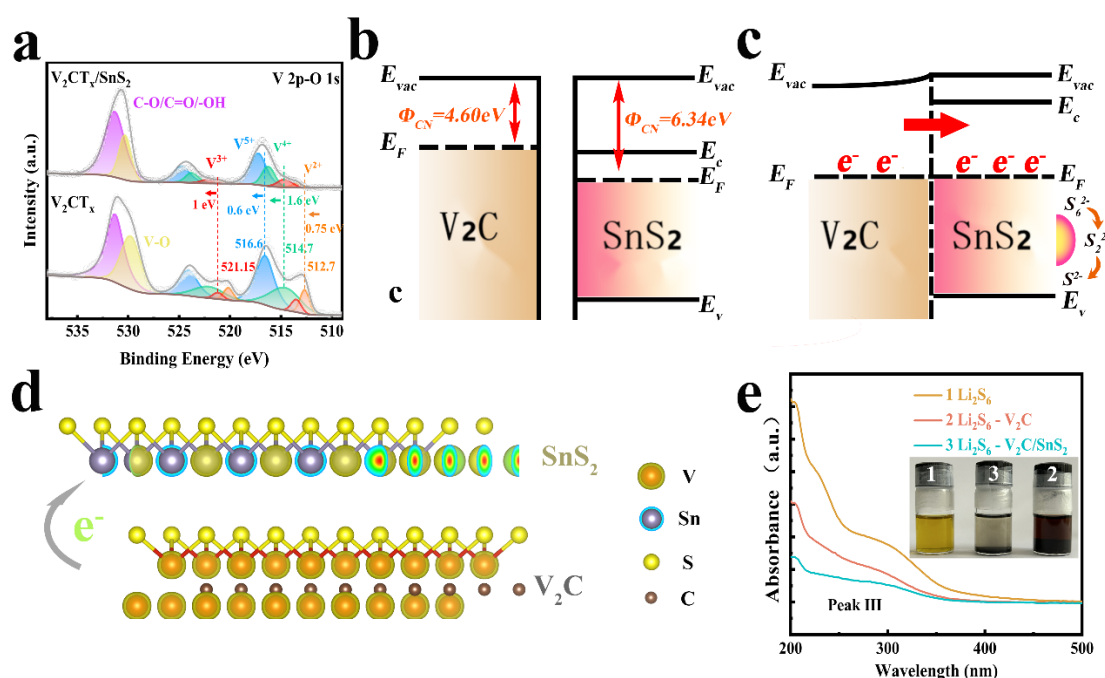


Figure 2. (a) V 2p XPS spectrum. (b, c) Energy-band diagrams before and after contact between V₂CT_x and SnS₂. (d) Optimized models of SnS₂ and the SnS₂-MXene heterostructure. (e) UV-Vis spectra and photographs after adsorption of Li₂S₆ by the blank, V₂CT_x and V₂CT_x/SnS₂ samples.

2.2. Electrochemical Performance of the Heterojunction

Cyclic voltammetry (CV) was used to compare Li–S cells with different separators (PP@V₂CT_x, PP@V₂CT_x/SnS₂, etc.). The PP@V₂CT_x/SnS₂ cell shows the strongest current response and the narrowest redox overpotential (Figure S9), especially at a scan rate of 0.1 mV s⁻¹, where the redox peak currents are notably higher than those of control cells. The first three CV cycles of PP@V₂CT_x/SnS₂ nearly overlap (Figure S10), indicating high reversibility and reduced polarization, suggesting that the PP@V₂CT_x/SnS₂ separator provides favorable kinetics for Li–S batteries.

Further analysis shows that the redox peaks of PP@V₂CT_x/SnS₂ are sharp and well-defined with low polarization (127 mV), evidencing pronounced catalytic activity in promoting LiPS redox reactions. This enhancement is attributed to the synergistic interaction between SnS₂ and MXene within the heterojunction, which improves LiPS conversion efficiency and reversibility.

Figure S11 shows that PP@V₂CT_x/SnS₂ exhibits the lowest polarization voltage ($\Delta E_1 = 127$ mV). The contribution of the two discharge plateaus ($\Delta Q_2/\Delta Q_1$) can quantify catalytic activity, where Q_1 corresponds to $S_8 \rightarrow Li_2S_4$ and Q_2 corresponds to $Li_2S_4 \rightarrow Li_2S$.^[30] A higher Q_2/Q_1 indicates stronger catalysis; the PP@V₂CT_x/SnS₂ cell achieves the highest Q_2/Q_1 (Figure 3a), demonstrating enhanced solid–liquid conversion catalysis.

CV at different scan rates was used to evaluate Li⁺ diffusion. As shown in Figure 3b and Figure S12. As scan rate increases, peak currents rise linearly with the square root of scan rate, indicating diffusion-controlled behavior.^[31] The PP@V₂CT_x/SnS₂ cell shows the highest peak currents and lowest polarization across 0.1–0.5 mV s⁻¹, and the largest Li⁺ diffusion coefficients for peaks I–III, suggesting accelerated LiPS conversion kinetics during galvanostatic cycling. Although accumulation of soluble LiPSs can increase electrolyte sulfur

concentration and viscosity—hindering Li^+ diffusion—the $\text{PP@V}_2\text{CT}_x/\text{SnS}_2$ cell still presents the highest DLi^+ . This is ascribed to the BIEF-assisted synergy in the heterojunction, which enhances chemical trapping of LiPSs and suppresses diffusion into the bulk electrolyte. Specifically, V_2CT_x provides a Li-friendly, high-adsorption-energy pathway for electron transport, while SnS_2 catalyzes fast redox reactions in the Li–S conversion cycle. Efficient electron transfer may also promote uniform Li nucleation and deposition on the anode, suppressing dendrite growth and extending cycle life. According to the Randles-Sevcik equation, the lithium ion diffusion coefficient can be derived from the slope of the $I_{\text{peak}}-\nu^{1/2}$ curve (where I_{peak} is the peak current and $\nu^{1/2}$ is the square root of the scan rate). The Figure S13 presents the linear fitting curves of the peak current of Peak I, Peak II, and Peak III against the square root of the scan rate. A comparison reveals that the slopes for Peak I (406.85), Peak II (179.86), and Peak III (199.54) of the $\text{PP@V}_2\text{CT}_x/\text{SnS}_2$ battery are significantly higher than those of the other two batteries. This further confirms that $\text{V}_2\text{CT}_x/\text{SnS}_2$ has excellent adsorption and catalytic activity, which can accelerate the migration of Li^+ and promote the conversion of LiPSs during the charge-discharge process. The CV results show that the current density of $\text{V}_2\text{CT}_x/\text{SnS}_2$ is significantly higher than that of V_2CT_x and SnS_2 , improving the conversion efficiency of LiPSs during the charge-discharge process. Li_2S nucleation and dissolution experiments further prove that $\text{V}_2\text{CT}_x/\text{SnS}_2$ significantly lowers the nucleation potential of Li_2S . Due to the strong catalytic effect and abundant active sites of SnS_2 , liquid-phase LiPSs are more easily nucleated by the $\text{V}_2\text{CT}_x/\text{SnS}_2$ material, and similar catalytic effects are observed during the oxidation of Li_2S , indicating that $\text{V}_2\text{CT}_x/\text{SnS}_2$ acts as a bifunctional catalyst.

Symmetric cells with Li_2S_6 electrolyte were assembled to evaluate electrocatalytic activity of $\text{V}_2\text{CT}_x/\text{SnS}_2$, V_2CT_x , and SnS_2 . At 10 mV s^{-1} (Figure 3c), the $\text{V}_2\text{CT}_x/\text{SnS}_2$ electrode shows two distinct redox peaks and the steepest peak currents, indicating enhanced LiPS conversion efficiency and electrochemical reversibility. These results suggest that dynamic adsorption by V_2CT_x facilitates accelerated redox reactions on SnS_2 surfaces.

To further investigate liquid–solid interfacial kinetics and heterostructure stability, V_2CT_x and $\text{V}_2\text{CT}_x/\text{SnS}_2$ were loaded onto carbon paper to study Li_2S nucleation and dissolution. Because V_2CT_x has a relatively low surface area and fewer nucleation sites, its nucleation current and capability are limited (Figure 3d). In contrast, $\text{V}_2\text{CT}_x/\text{SnS}_2$ significantly lowers the Li_2S nucleation overpotential, indicating that liquid LiPSs nucleate more readily on $\text{V}_2\text{CT}_x/\text{SnS}_2$ due to the catalytic heterointerface. Moreover, $\text{V}_2\text{CT}_x/\text{SnS}_2$ shows much faster Li_2S dissolution ($2749 \text{ s} / 810.56 \text{ mAh g}^{-1}$) than V_2CT_x ($4114 \text{ s} / 584.28 \text{ mAh g}^{-1}$) (Figure 3e), demonstrating that the heterojunction also accelerates sulfur species conversion during oxidation. Thus, the $\text{V}_2\text{CT}_x/\text{SnS}_2$ heterojunction promotes both Li_2S nucleation and oxidative dissolution, providing reversible bifunctional catalysis that is particularly important for ultrafast-charging Li–S batteries.

Cycling stability at different rates was evaluated. Figure S14 shows that, compared with PP (771.2 mAh g^{-1}) and $\text{PP}@V_2\text{CT}_x$ (687.5 mAh g^{-1}), $\text{PP}@V_2\text{CT}_x/\text{SnS}_2$ delivers a higher initial discharge capacity (936.7 mAh g^{-1}) and maintains superior capacity after 200 cycles at 0.5 C . Rate performance comparisons at 0.2 , 0.5 , 1 , 2 , and 4 C (Figure 3f) show capacities of 1256.1 , 995.1 , 863.9 , 763.6 , and 653.0 mAh g^{-1} for $\text{PP}@V_2\text{CT}_x/\text{SnS}_2$. Notably, when the current returns to 0.2 C , a reversible capacity of 981.6 mAh g^{-1} is retained. Even at 4 C , two

discharge plateaus remain clearly distinguishable (Figure S15), indicating fast conversion kinetics and structural stability of the heterojunction under high-rate operation.

To further elucidate the role of the V_2CT_x/SnS_2 interlayer in interfacial charge transfer and LiPSs adsorption, electrochemical impedance spectroscopy (EIS) was performed on batteries assembled with different separators, including both fresh cells and those cycled for 1000 cycles (1 C). Figure S16 shows that the diameter of the resistance semicircle for the $PP@V_2CT_x/SnS_2$ battery is only 30 Ω , significantly smaller than that of $PP@V_2CT_x$ (55 Ω) and PP (65 Ω), indicating that the internal resistance of the $PP@V_2CT_x/SnS_2$ battery is the lowest. Additionally, the straight line after the semicircle has the largest slope, reflecting faster charge transport within the battery, which is related to the re-dispersion of the electrolyte and LiPSs in the electrode. Compared to the PP and $PP@V_2CT_x$ cells, the $PP@V_2CT_x/SnS_2$ cells show the smallest R_{ct} semicircle in both fresh and cycled cells, suggesting that the V_2CT_x/SnS_2 interlayer facilitates rapid electron transfer and fast redox kinetics.

Long-term cycling (Figure S17) shows that $PP@V_2CT_x/SnS_2$ maintains excellent durability, delivering 430.3 mAh g⁻¹ after 1000 cycles at 1.0 C with 55.2% capacity retention, outperforming PP and $PP@Ti_3C_2T_x$. At an ultrahigh rate of 2 C, $PP@V_2CT_x/SnS_2$ sustains 3000 cycles with an average capacity decay of only 0.019% per cycle and Coulombic efficiency above 99% (Figure 3g). This outstanding performance is attributed to accelerated redox reactions and high sulfur utilization enabled by the heterojunction-induced BIEF. Based on the above results, the $PP@V_2CT_x/SnS_2$ separator shows strong competitiveness compared to reported MXene heterojunction-modified separator batteries and other outstanding works (Supplementary Table 1).

Post-cycling SEM of the $V_2CT_x/SnS_2/PP$ separator (Figure S18) shows that the modified layer remains tightly adhered to the PP separator after prolonged cycling, with no severe surface damage from LiPS corrosion, indicating compositional stability after LiPS adsorption. (Figure 3h).

Figure 3h provides compelling evidence of the contribution of the Ohmic heterojunction as a super catalyst and super adsorbent in enhancing battery performance, demonstrating that the combination of V_2CT_x and SnS_2 improves the redox reactions of sulfur, mitigates the shuttle effect of polysulfides, and enhances the overall performance of the lithium-sulfur battery.

Notably, it contributes to both the ultra-long cycle life and high rate performance.

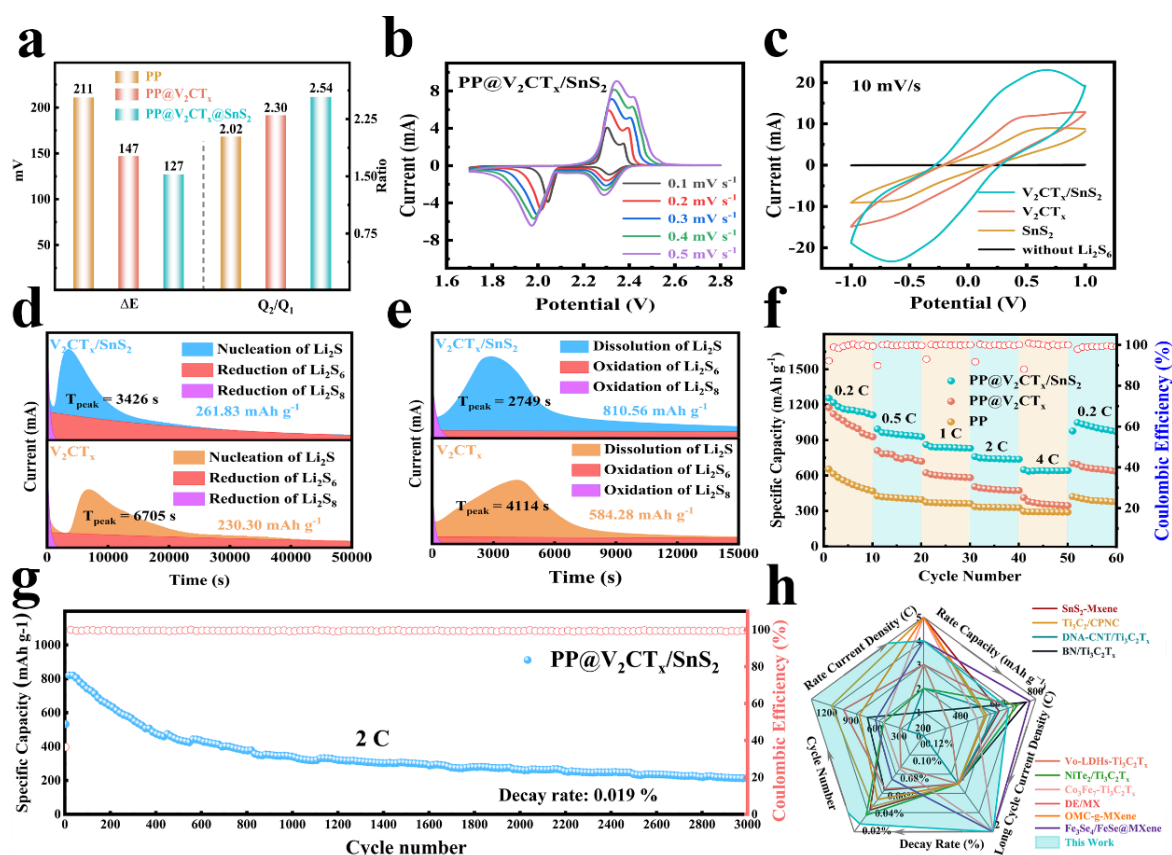


Figure 3. (a) ΔE and Q_2/Q_1 derived from galvanostatic charge–discharge curves. (b) CV curves of $PP@V_2CT_x/SnS_2$ at 0.1–0.5 $mV s^{-1}$. (c) CV curves of symmetric cells at 10 $mV s^{-1}$. (d) Li_2S precipitation and (e) dissolution profiles on V_2CT_x/SnS_2 and V_2CT_x electrodes. (f) Rate performance. (g) Long-term cycling at 2 C. (h) Comparison with reported composite separators.

3. Catalytic Mechanism of the SnS_2 –MXene Heterostructure

To further elucidate the merits of the heterojunction, we analyzed mechanistic changes induced by the composite. Restacking of V_2CT_x reduces active-site density and weakens LiPS adsorption, increasing the probability of LiPSs crossing the separator and thereby deteriorating battery performance. Formation of the composite effectively mitigates V_2CT_x restacking and also suppresses SnS_2 aggregation. SnS_2 exhibits strong catalytic activity for sulfur redox conversion; its trilayer-stacked atomic arrangement connected by van der Waals interactions favors rapid charge diffusion and transfer^[32]. Concurrently, V_2CT_x provides efficient electron-conducting pathways, compensating for SnS_2 in conductivity.

The strong adsorption and large surface area of V_2CT_x provide abundant active sites for LiPSs, facilitating electrochemical reactions and improving charge-storage capability and ionic adsorption efficiency. V_2CT_x displays amphiphilic behavior toward LiPSs and offers both adsorption and catalytic functionalities, making it an effective multifunctional interfacial material. Under the synergistic effect of SnS_2 catalysis and the high conductivity and strong adsorption of V_2CT_x , electron/ion transport across the interface is significantly enhanced.

Together with the BIEF induced by the heterojunction, a heterointerface with “super adsorption” and “super catalysis” is formed. The rectifying effect of the BIEF drives directional electron transfer from the V_2CT_x side to the SnS_2 side, generating electron-depleted (oxidized) and electron-enriched (reduced) regions. This rectifying interface is crucial for efficient redox processes: electron enrichment on SnS_2 strengthens LiPS capture and accelerates LiPS reduction during discharge, while V_2CT_x facilitates LiPS oxidation during charge.

This mechanism ensures fast redox kinetics throughout cycling and provides new insights into enhancing redox performance via electron regulation in heterojunctions. Importantly, the V_2CT_x/SnS_2 -modified separator exhibits excellent mechanical stability, effectively buffering electrode volume changes associated with the S_8/Li_2S conversion and maintaining electrode structural integrity. Therefore, the synergistic effects originating from the V_2CT_x/SnS_2 heterojunction not only improve long-term cycling stability but also enhance high-rate capability, offering robust support for high-performance Li–S batteries.

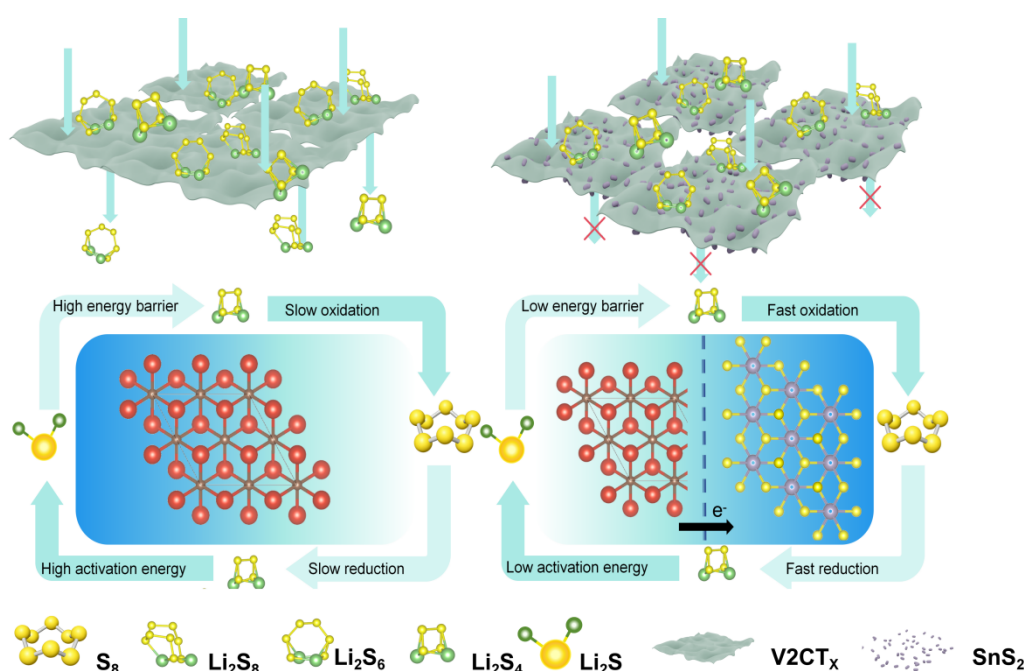


Figure 4. Schematic illustration of the Li–S battery equipped with a V_2CT_x/SnS_2 -modified separator and the proposed adsorption–catalysis mechanism.

4. Conclusions

This study presents the design of a V_2CT_x/SnS_2 heterostructure-modified separator, which significantly enhances the electrochemical performance of lithium-sulfur (Li–S) batteries. The combination of SnS_2 and V_2CT_x addresses key issues such as the LiPSs shuttle effect, slow sulfur conversion, and poor cycling stability. SnS_2 improves catalytic activity and promotes the conversion of LiPSs to Li_2S , while V_2CT_x offers excellent electronic conductivity and

strong LiPSs adsorption, effectively suppressing the shuttle effect and improving capacity utilization.

The experimental results show that the V_2CT_x/SnS_2 heterostructure separator outperforms traditional materials, achieving a first-cycle discharge capacity of $1256.1 \text{ mAh g}^{-1}$ at 0.2 C and maintaining excellent long-term cycling stability, with a capacity decay rate of only 0.019% after 3000 cycles. The battery also demonstrates strong performance under high-rate conditions.

In conclusion, the V_2CT_x/SnS_2 heterojunction enhances Li-S battery performance through a synergistic effect, addressing the shuttle effect, accelerating sulfur conversion, and improving overall cycling stability and rate performance. This research provides a promising approach for enhancing the performance of Li-S batteries and opens new possibilities for the application of heterostructure materials in energy storage.

5. Experimental Section

5.1. Preparation of V_2CT_x Nanosheets

V_2CT_x nanosheets were prepared via an etching process. First, 18 mL of 48% hydrofluoric acid (HF) and 12 mL of concentrated hydrochloric acid (12 M) were added into a large Teflon-lined vessel and stirred at 150 rpm with a Teflon magnetic stir bar. Then, 1.5 g of ice-bath-cooled MAX precursor (V_2AlC) was slowly introduced. The etching reaction was conducted at 50°C in a water bath at 400 rpm for 72 h. The resulting acidic mixture was centrifuged and washed repeatedly until the supernatant reached $\text{pH} \geq 5.5$. The sediment was collected and mixed with 10 mL of 10% tetrabutylammonium hydroxide (TBAOH) and 10 mL of deionized water, shaken thoroughly, and intercalated at 25°C (water bath, 400 rpm) for

12 h under inert atmosphere. After multiple centrifugation/washing steps, the supernatant (pH = 7) was collected and freeze-dried to obtain V_2CT_x powder.

5.2. Preparation of Ellipsoidal SnS_2

Ellipsoidal SnS_2 was synthesized as follows. First, 1.56 g of $SnCl_4 \cdot 5H_2O$ was dissolved in deionized water under stirring until fully dissolved. Then, 2 g of C_2H_5NS was added and stirred until complete dissolution. The solution was transferred into a 100 mL Teflon-lined autoclave and heated at 160 °C for 720 min. After cooling, the product was collected via repeated centrifugation/washing, and the solid was dried in a hot-air oven at 80 °C for 360 min to yield ellipsoidal SnS_2 .

5.3. Preparation of the V_2CT_x/SnS_2 Composite

The V_2CT_x/SnS_2 composite was prepared by a hydrothermal route. Specifically, 90 mg of V_2CT_x was dispersed in deionized water and ultrasonicated for 1 h in an ice-water bath under inert atmosphere to avoid oxidation. Separately, 156 mg of $SnCl_4 \cdot 5H_2O$ and 200 mg of C_2H_5NS were dissolved in another container and stirred until fully dissolved. The two solutions were then mixed and stirred for 20 min, transferred into a 100 mL Teflon-lined autoclave, and heated at 160 °C for 720 min. After reaction, the product was separated by repeated centrifugation/washing and freeze-dried to obtain V_2CT_x/SnS_2 black powder.

5.4. Static Adsorption Test and UV–Vis Spectroscopy

20 mg of V_2CT_x/SnS_2 or V_2CT_x was placed into a vial and mixed with 2 mL of 4 mM Li_2S_6 solution. After standing for 6 h, the supernatant was collected for UV–Vis measurements to evaluate LiPS adsorption.

5.5. Preparation of Functional Separators

V_2CT_x/SnS_2 functional separators were prepared by a surface-coating method. V_2CT_x/SnS_2 (40 mg), Super P Li (5 mg), and PVDF (5 mg) were mixed in NMP to form a slurry and coated

onto carbon paper. The coated film was dried under vacuum at 60 °C for 12 h and punched into 12 mm disks. The areal loading of active material was controlled at $\sim 0.5 \text{ mg cm}^{-2}$.

Symmetric cells were assembled by adding 20 μL of electrolyte (0.5 M Li_2S_6 and 1 M LiTFSI in DOL/DME, 1:1 v/v) to both sides of the separator. The assembled cells were rested for 5 h. V_2CT_x and SnS_2 symmetric cells were prepared similarly.

5.6. Preparation of the Li_2S_8 Electrolyte

Li_2S_8 electrolyte was prepared by dissolving S_8 and Li_2S in tetraethylene glycol dimethyl ether at a molar ratio of 7:1, followed by addition of 1 M LiTFSI and stirring at 60 °C for 24 h. For testing, 20 μL of 0.5 M Li_2S_8 solution was used as the catholyte, and 20 μL of Li_2S_8 -free electrolyte was used as the anolyte. $\text{V}_2\text{CT}_x/\text{SnS}_2$ or V_2CT_x served as the electrode and lithium foil as the counter electrode. Cells were rested for 5 h prior to measurements to ensure full wetting.

5.7. Preparation of the KB@S Cathode

KB@S was prepared via a melt-diffusion process. Briefly, 150 mg of KB-600JD and 350 mg sublimed sulfur were mixed and ground, then heated at 155 °C for 24 h under Ar. The cathode slurry was prepared by grinding KB@S (80 mg), PVDF (10 mg), and Super P Li (10 mg) in NMP and coating onto carbon-coated Al foil, followed by vacuum drying at 60 °C for 12 h. The cathode was punched into 12 mm disks with a sulfur loading of $\sim 1.1 \text{ mg cm}^{-2}$.

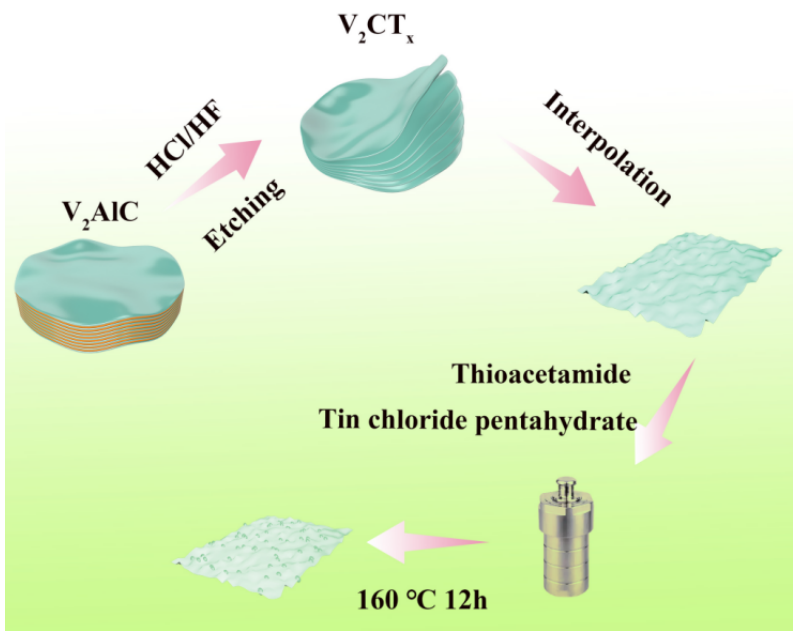


Figure 5. Schematic illustration of the $\text{V}_2\text{CT}_x/\text{SnS}_2$ synthesis route.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

All data necessary to support the conclusions of this study are included in the main text or the Supporting Information.

Acknowledgements

This work was supported by the Natural Science Foundation of Shandong Province, China (Grant No. ZR2024MF089); the Key R&D Program of Shandong Province (Competitive Innovation Platform) (Grant No. 2026CXPT230); and the Major Basic Research Program of the Natural Science Foundation of Shandong Province (Grant No. ZR2026ZD47).

References

1. P. Bonnick and J. Muldoon, “The Dr Jekyll and Mr Hyde of lithium sulfur batteries,” *Energy & Environmental Science* 13, no. 12 (2020): 4808–4833.
<https://doi.org/10.1039/d0ee02797a>

2. J. Jiang, J. Ontaneda, S. Pal, Z. Guo, C. Forrester, K. Zheng, M. Wang, J. Briscoe, M.-M. Titirici, and H. Au, “Enhanced polysulfide trapping in Li–S batteries by dipole alignment in ferroelectric BaTiO₃,” *Energy & Environmental Science* 17, no. 17 (2024): 6291–6301.
<https://doi.org/10.1039/d4ee01936a>
3. S.-H. Yu, X. Huang, K. Schwarz, R. Huang, T.A. Arias, J.D. Brock, and H.D. Abruña, “Direct visualization of sulfur cathodes: new insights into Li–S batteries via operando X-ray based methods,” *Energy & Environmental Science* 11, no. 1 (2018): 202–210.
<https://doi.org/10.1039/c7ee02874a>
4. S. Rehman, M. Pope, S. Tao, and E. McCalla, “Evaluating the effectiveness of in situ characterization techniques in overcoming mechanistic limitations in lithium–sulfur batteries,” *Energy & Environmental Science* 15, no. 4 (2022): 1423–1460.
<https://doi.org/10.1039/d1ee03396d>
5. W. Zuo, M. Luo, X. Liu, J. Wu, H. Liu, J. Li, M. Winter, R. Fu, W. Yang, and Y. Yang, “Li-rich cathodes for rechargeable Li-based batteries: reaction mechanisms and advanced characterization techniques,” *Energy & Environmental Science* 13, no. 12 (2020): 4450–4497. <https://doi.org/10.1039/d0ee01694b>
6. M. Wild, L. O'Neill, T. Zhang, R. Purkayastha, G. Minton, M. Marinescu, and G.J. Offer, “Lithium sulfur batteries, a mechanistic review,” *Energy & Environmental Science* 8, no. 12 (2015): 3477–3494. <https://doi.org/10.1039/c5ee01388g>
7. A. Kim, S.H. Oh, A. Adhikari, B.R. Sathe, S. Kumar, and R. Patel, “Recent advances in modified commercial separators for lithium–sulfur batteries,” *Journal of Materials Chemistry A* 11, no. 15 (2023): 7833–7866. <https://doi.org/10.1039/d2ta09266b>

8. J. Tang, Q. Zhao, F. Li, Z. Hao, X. Xu, Q. Zhang, J. Liu, Y. Jin, and H. Wang, “Two-dimensional materials towards separator functionalization in advanced Li–S batteries,” *Nanoscale* 13, no. 45 (2021): 18883–18911. <https://doi.org/10.1039/d1nr05489a>
9. M. Wu, Y. He, L. Wang, Q. Xia, and A. Zhou, “Synthesis and electrochemical properties of V₂C MXene by etching in opened/closed environments,” *Journal of Advanced Ceramics* 9, no. 6 (2020): 749–758. <https://doi.org/10.1007/s40145-020-0411-8>
10. Z. Wang, G. Tan, B. Zhang, Q. Yang, S. Feng, Y. Liu, T. Liu, L. Guo, C. Zeng, W. Liu, et al., “Intrinsic Polarized Electric Field Induces a Storing Mechanism to Achieve Energy Storing Catalysis in V₂C MXene,” *Advanced Materials* 36, no. 3 (2024): 2307795. <https://doi.org/10.1002/adma.202307795>
11. D. Sabaghi, J. Polčák, H. Yang, X. Li, A. Morag, D. Li, A.S. Nia, S. Khosravi H, T. Šikola, X. Feng, et al., “Multifunctional Molecule-Grafted V₂C MXene as High-Kinetics Potassium-Ion-Intercalation Anodes for Dual-Ion Energy Storage Devices,” *Advanced Energy Materials* 14, no. 3 (2024): 2302961. <https://doi.org/10.1002/aenm.202302961>
12. Y. Cao, H. Chen, Y. Shen, M. Chen, Y. Zhang, L. Zhang, Q. Wang, S. Guo, and H. Yang, “SnS₂ Nanosheets Anchored on Nitrogen and Sulfur Co-Doped MXene Sheets for High-Performance Potassium-Ion Batteries,” *ACS Applied Materials & Interfaces* 13, no. 15 (2021): 17668–17676. <https://doi.org/10.1021/acsami.1c02590>
13. J. Li, L. Han, Y. Li, J. Li, G. Zhu, X. Zhang, T. Lu, and L. Pan, “MXene-decorated SnS₂/Sn₃S₄ hybrid as anode material for high-rate lithium-ion batteries,” *Chemical Engineering Journal* 380 (2020): 122590. <https://doi.org/10.1016/j.cej.2019.122590>
14. L. Luo, N. Shi, D. Lu, C. Xu, J. Zhang, M.Z. Hasan, R. Zhang, G. Cai, S. Sha, and J. Zhu, “Unveiling the mechanism of in situ construction of heterostructured

- SnS₂@MXene/porous carbon nanofibers enabled superior lithium-ion batteries,” *Journal of Colloid and Interface Science* 704 (2026): 139407. <https://doi.org/10.1016/j.jcis.2025.139407>
15. Y. Wu, P. Nie, L. Wu, H. Dou, and X. Zhang, “2D MXene/SnS₂ composites as high-performance anodes for sodium ion batteries,” *Chemical Engineering Journal* 334 (2018): 932–938. <https://doi.org/10.1016/j.cej.2017.10.007>
 16. W. Wang, J. Wu, X. Chen, X. Shen, X. Jin, Y. Sun, C. Yan, Y. Li, and P. Zhang, “MXene-Induced Construction of SnS₂ Nano-Arrays with Sulfur Vacancies for High-Efficiency Photocatalytic CO₂ Reduction,” *Chemistry – An Asian Journal* 20, no. 10 (2025): e202500086. <https://doi.org/10.1002/asia.202500086>
 17. T. He, S. Sun, B. Huang, and X. Li, “MXene/SnS₂ Heterojunction for Detecting Sub-ppm NH₃ at Room Temperature,” *ACS Applied Materials & Interfaces* 15, no. 3 (2023): 4194–4207. <https://doi.org/10.1021/acsami.2c18097>
 18. Q. Zhao, X. Bao, L. Meng, S. Dong, Y. Zhang, C. Qing, T. Zhu, and H.-E. Wang, “Nitrogen-doped hollow carbon@tin disulfide as a bipolar dynamic host for lithium-sulfur batteries with enhanced kinetics and cyclability,” *Journal of Colloid and Interface Science* 644 (2023): 546–555. <https://doi.org/10.1016/j.jcis.2023.03.169>
 19. S. Liu, C. Zhang, W. Yue, X. Chen, and X. Yang, “Graphene-Based Mesoporous SnO₂ Nanosheets as Multifunctional Hosts for High-Performance Lithium–Sulfur Batteries,” *ACS Applied Energy Materials* 2, no. 7 (2019): 5009–5018. <https://doi.org/10.1021/acsaem.9b00680>
 20. J. Deng, G. Du, Y. Wang, H. Li, D. Han, S. Chen, Y. Wang, W. Zhao, S. Ding, Q. Su, et al., “Hollow ZnS-SnS₂@MoS₂ heterostructures for high-efficiency adsorption-storage-

catalysis of polysulfides in lithium-sulfur batteries,” *Journal of Alloys and Compounds* 1010

(2025): 177172. <https://doi.org/10.1016/j.jallcom.2024.177172>

21. V.P. Letswalo, T.A. Makhetha, M.P. Seopela, A. Szymczyk, and S.P. Malinga,

“Fabrication of a novel tin disulfide/vanadium carbide MXene (SnS₂/V₂CT_x) photocatalytic heterojunction with the potential application for environmental remediation,” *Materials Science in Semiconductor Processing* 184 (2024): 108803.

Science in Semiconductor Processing 184 (2024): 108803.

<https://doi.org/10.1016/j.mssp.2024.108803>

22. Y. Guan, S. Jiang, Y. Cong, J. Wang, Z. Dong, Q. Zhang, G. Yuan, Y. Li, and X. Li,

“A hydrofluoric acid-free synthesis of 2D vanadium carbide (V₂C) MXene for supercapacitor electrodes,” *2D Materials* 7, no. 2 (2020): 025010. <https://doi.org/10.1088/2053-1583/ab6706>

23. R.A. Beers, D. Bao, and J.R. Ray, “Chemical Intercalant Affects the Structural Properties and Aqueous Stability of V₂CT_x MXene,” *Advanced Materials Interfaces* 12, no.

14 (2025): e00145. <https://doi.org/10.1002/admi.202500145>

24. C. Wan, L. Zhou, L. Sun, L. Xu, D.-G. Cheng, F. Chen, X. Zhan, and Y. Yang,

“Boosting visible-light-driven hydrogen evolution from formic acid over AgPd/2D g-C₃N₄ nanosheets Mott-Schottky photocatalyst,” *Chemical Engineering Journal* 396 (2020):

125229. <https://doi.org/10.1016/j.cej.2020.125229>

25. H. Wang, Y. Wei, G. Wang, Y. Pu, L. Yuan, C. Liu, Q. Wang, Y. Zhang, and H. Wu,

“Selective Nitridation Crafted a High-Density, Carbon-Free Heterostructure Host with Built-In Electric Field for Enhanced Energy Density Li–S Batteries,” *Advanced Science* 9, no. 23

(2022): 2201823. <https://doi.org/10.1002/advs.202201823>

26. J. Halim, K.M. Cook, M. Naguib, P. Eklund, Y. Gogotsi, J. Rosen, and M.W.

Barsoum, “X-ray photoelectron spectroscopy of select multi-layered transition metal carbides

(MXenes),” *Applied Surface Science* 362 (2016): 406–417.

<https://doi.org/10.1016/j.apsusc.2015.11.089>

27. X. Wang, J. Qiao, L. Guo, M. Feng, P. Wang, and Z. Liu, “Constructing Built-in Electric Field in Heterogeneous Flower-Like SnS₂@Few-Layer Ti₃C₂ for Enhanced Charge Transfer Kinetics in Lithium-ion Capacitors,” *Advanced Functional Materials* 35, no. 32 (2025): 2422017. <https://doi.org/10.1002/adfm.202422017>
28. L. Wang, G.-R. Li, S. Liu, and X.-P. Gao, “Hollow Molybdate Microspheres as Catalytic Hosts for Enhancing the Electrochemical Performance of Sulfur Cathode under High Sulfur Loading and Lean Electrolyte,” *Advanced Functional Materials* 31, no. 18 (2021): 2010693. <https://doi.org/10.1002/adfm.202010693>
29. R. Liu, Z. Wei, L. Peng, L. Zhang, A. Zohar, R. Schoeppner, P. Wang, C. Wan, D. Zhu, H. Liu, et al., “Establishing reaction networks in the 16-electron sulfur reduction reaction,” *Nature* 626, no. 7997 (2024): 98–104. <https://doi.org/10.1038/s41586-023-06918-4>
30. C. Zhang, R. Du, J.J. Biendicho, M. Yi, K. Xiao, D. Yang, T. Zhang, X. Wang, J. Arbiol, J. Llorca, et al., “Tubular CoFeP@CN as a Mott–Schottky Catalyst with Multiple Adsorption Sites for Robust Lithium–Sulfur Batteries,” *Advanced Energy Materials* 11, no. 24 (2021): 2100432. <https://doi.org/10.1002/aenm.202100432>
31. D. Yang, M. Li, X. Zheng, X. Han, C. Zhang, J. Jacas Biendicho, J. Llorca, J. Wang, H. Hao, J. Li, et al., “Phase Engineering of Defective Copper Selenide toward Robust Lithium–Sulfur Batteries,” *ACS Nano* 16, no. 7 (2022): 11102–11114. <https://doi.org/10.1021/acsnano.2c03788>
32. L. Chen, L. Yue, X. Wang, S. Wu, W. Wang, D. Lu, X. Liu, W. Zhou, and Y. Li, “Synergistically Accelerating Adsorption-Electrocatalysis of Sulfur Species via Interfacial

WILEY-VCH

Built-In Electric Field of SnS₂-MXene Mott–Schottky Heterojunction in Li-S Batteries,”

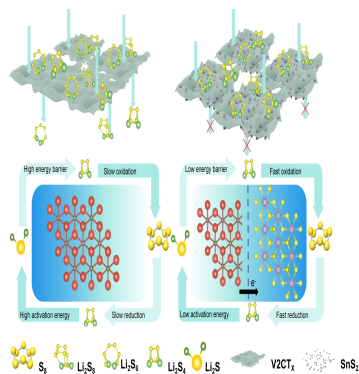
Small 19, no. 15 (2023): 2206462. <https://doi.org/10.1002/sml.202206462>

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Table of Contents

A V_2CT_x/SnS_2 heterostructure is engineered as a multifunctional separator coating for lithium–sulfur batteries. Conductive V_2CT_x nanosheets accelerate electron transport, while polar SnS_2 provides strong polysulfide adsorption and catalytic sites. Interfacial electronic coupling creates a built-in electric field that promotes bidirectional sulfur redox reactions, suppresses polysulfide shuttling, and enables durable, high-rate battery performance.



Supporting Information

Dual-Function Catalytic Regulation of Polysulfides by V_2CT_x/SnS_2 Heterostructures Enabling High-Performance Li–S Battery Separators

Longyu Liu^{a,b,+,}, Fei Xing^{a,b,*,+}, Songshe Xu^b, Yingfei Wen^b, Minghui Dong^c, Qi Fan^c, Ze Zhang^{a,d,e,f,*}

^aCollege of Photonics and Optical Engineering, Aerospace Information Technology University, Jinan, China.

^bSchool of Physics and Optoelectronic Engineering, Shandong University of Technology, Zibo, China.

^cSchool of Materials Science and Engineering, Southeast University, Nanjing, China.

^dAerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

^eKey Laboratory of Computational Optical Imaging Technology, Chinese Academy of Sciences, Beijing, China.

^fQilu Aerospace Information Research Institute, Jinan, China.

*E-mail: Fei Xing(xingfei@sdut.edu.cn) and Ze Zhang(zhangze@aircas.ac.cn)

⁺Longyu Liu and Fei Xing contributed equally to this work.

S1. Experimental and Computational Details**S1.1 Scanning Electron Microscopy (SEM)**

SEM images were acquired using a Hitachi SU8010 cold-field-emission scanning electron microscope. Samples were mounted on conductive adhesive on a specimen holder and sputter-coated with gold before imaging.

S1.2 Transmission Electron Microscopy (TEM)

High-resolution TEM images were acquired using a Philips Tecnai F20 instrument. Samples were dispersed in anhydrous ethanol by ultrasonication, deposited onto an ultrathin support grid, and dried before characterization.

S1.3 X-Ray Diffraction (XRD)

XRD patterns were collected using a Rigaku MiniFlex 600 powder diffractometer with Cu K α radiation ($\lambda = 1.54060 \text{ \AA}$). Data were recorded over a 2θ range of $5\text{--}80^\circ$ at a scan rate of $10^\circ \text{ min}^{-1}$.

S1.4 Ultraviolet–Visible (UV–Vis) Absorption Spectroscopy

A Thermo Fisher UV–Vis spectrophotometer was used to evaluate the residual polysulfide concentration after adsorption by the different materials. Spectra were recorded over 300–550 nm.

S1.5 X-Ray Photoelectron Spectroscopy (XPS)

XPS measurements were performed using a Thermo Scientific K-Alpha instrument with Al K α radiation ($h\nu = 1486.6 \text{ eV}$) and a $400 \text{ }\mu\text{m}$ beam. Binding energies were calibrated to the C 1s peak at 284.8 eV , and peak fitting was performed using Avantage software.

S1.6 First-Principles Calculations

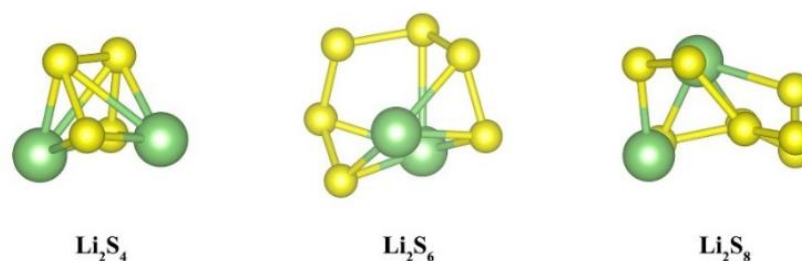
All calculations were performed within density functional theory using the projector augmented-wave method, as implemented in the Vienna Ab initio Simulation Package (VASP).^[1] The Perdew–Burke–Ernzerhof generalized-gradient approximation was used for the exchange–correlation functional.^[2] Long-range van der Waals interactions were described using the DFT-D3 method.^[3]

The plane-wave cutoff energy was 500 eV, and the electronic self-consistency criterion was 10^{-5} eV. A 15 Å vacuum layer was introduced perpendicular to the sheet to minimize interactions between periodic images. Brillouin-zone integration employed a $1 \times 1 \times 1$ k-point mesh. All structures were relaxed until the residual force on each atom was below 0.02 eV Å^{-1} .

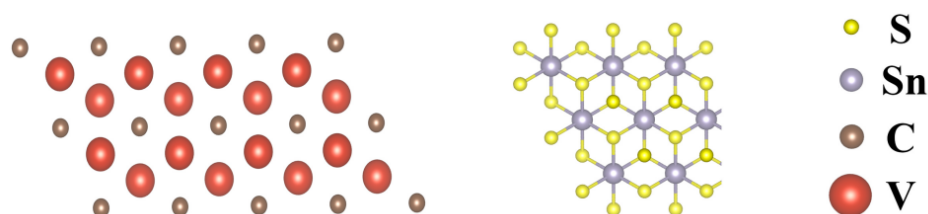
The adsorption energy (E_{ads}) was calculated using Equation S1:

$$E_{\text{ads}} = E_{\text{total}} - E_{\text{sub}} - E_{\text{adso}} \quad (\text{S1})$$

where E_{total} is the total energy of the optimized adsorbate–substrate system, and E_{sub} and E_{adso} are the energies of the isolated substrate and adsorbate, respectively, calculated in the same supercell.



Scheme S1. Computational models of the ground-state Li_2S_4 , Li_2S_6 , and Li_2S_8 clusters.



Scheme S2. Top views of the computational models of (a) V_2C and (b) SnS_2 .

S1.7 Coin-Cell Assembly

CR2032 coin cells were assembled in an argon-filled glovebox with H_2O and O_2 concentrations below 0.1 ppm. Lithium-metal foil, a KB@S cathode, and a pristine or modified polypropylene separator (Celgard 2500) were used. The electrolyte consisted of 1.0 M lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) and 2 wt% anhydrous LiNO_3 dissolved in 1,3-dioxolane/1,2-dimethoxyethane (DOL/DME; 1:1, v/v). A total of 40 μL of electrolyte was used per cell (20 μL on each side of the separator). The sulfur loading was approximately 1.1 mg cm^{-2} . The assembly sequence was negative case, lithium-metal foil, electrolyte, separator, electrolyte, cathode, spacer, spring, and positive case, followed by pressure sealing. Cells were rested before testing, and electrochemical measurements were performed at $25 \pm 2^\circ\text{C}$.

S1.8 Cyclic Voltammetry (CV)

CV measurements of Li–S cells and symmetric cells were performed using an IVIUM electrochemical workstation. Li–S cells were scanned from 1.7 to 2.8 V at $0.1\text{--}0.5 \text{ mV s}^{-1}$. Symmetric cells were scanned from -1.0 to 1.0 V at 10 mV s^{-1} .

S1.9 Electrochemical Impedance Spectroscopy (EIS)

EIS measurements were conducted using an IVIUM Vertex.C electrochemical workstation over a frequency range of 100 kHz–0.1 Hz with a perturbation amplitude of 5 mV.

S1.10 Galvanostatic Charge–Discharge Testing

Galvanostatic charge–discharge measurements were performed using a Neware CT-4008Tn battery-testing system over 1.7–2.8 V. A specific current of $1\text{ C} = 1675\text{ mA g}^{-1}$ was used for rate conversion. The reported metrics include specific capacity, Coulombic efficiency and capacity-decay rate.

S2. Supporting Figures

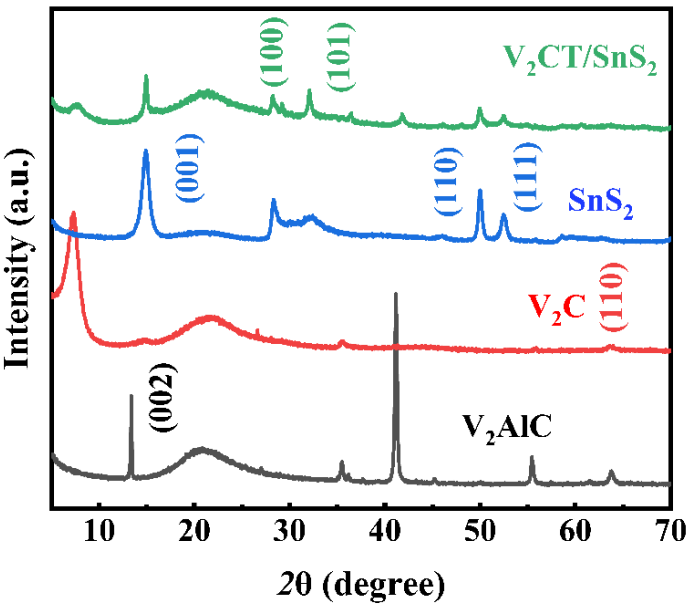


Figure S1. XRD patterns of V₂AlC, V₂C, SnS₂ and the V₂CT_x/SnS₂ composite.

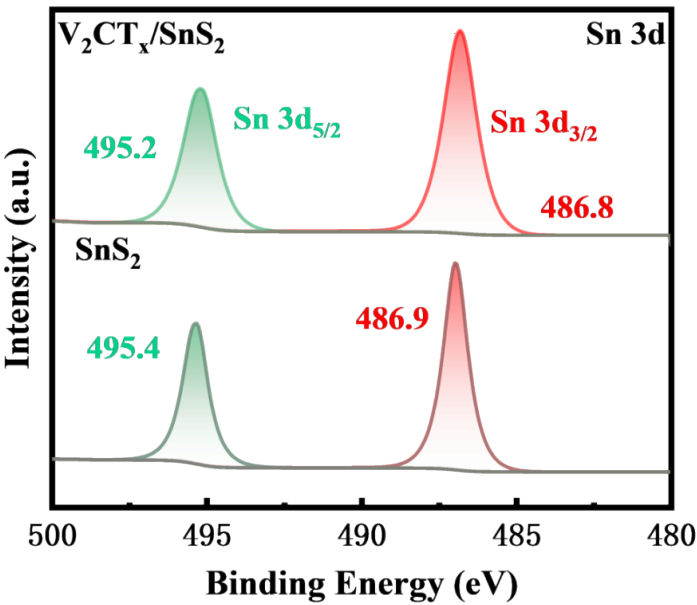


Figure S2. High-resolution Sn 3d XPS spectra of SnS₂ and V₂CT_x/SnS₂.

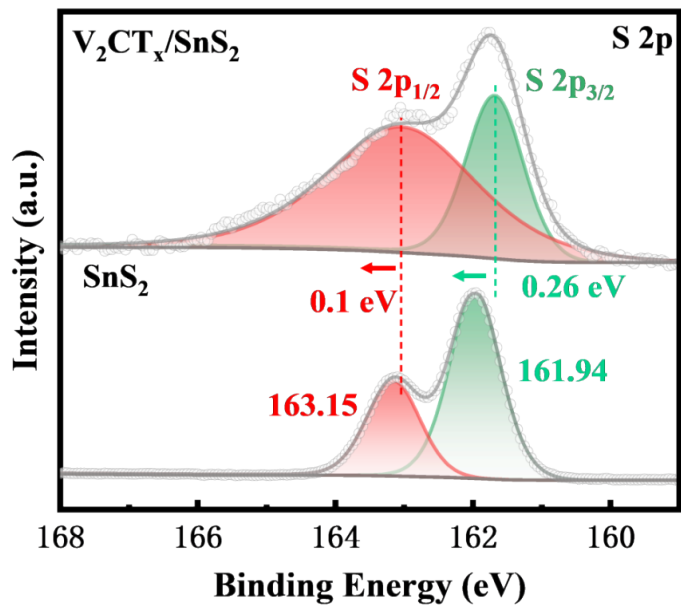


Figure S3. High-resolution S 2p XPS spectra of SnS_2 and V_2CT_x/SnS_2 .

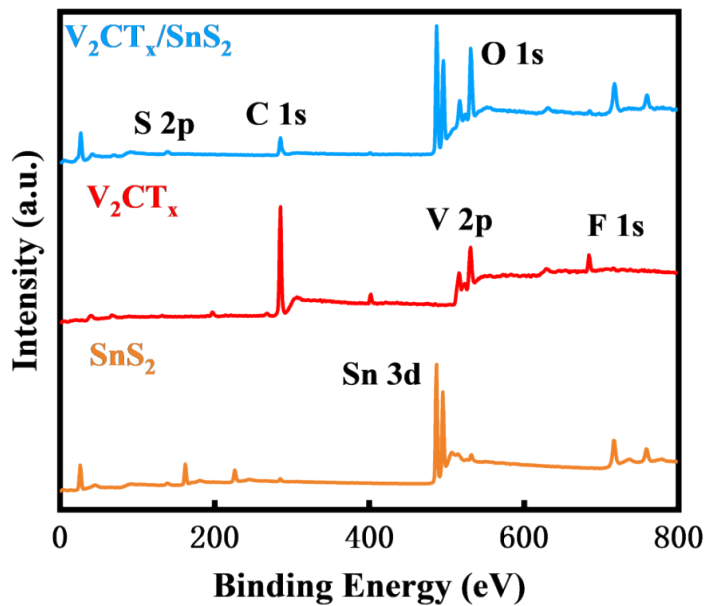


Figure S4. XPS survey spectra of SnS_2 , V_2CT_x and V_2CT_x/SnS_2 .

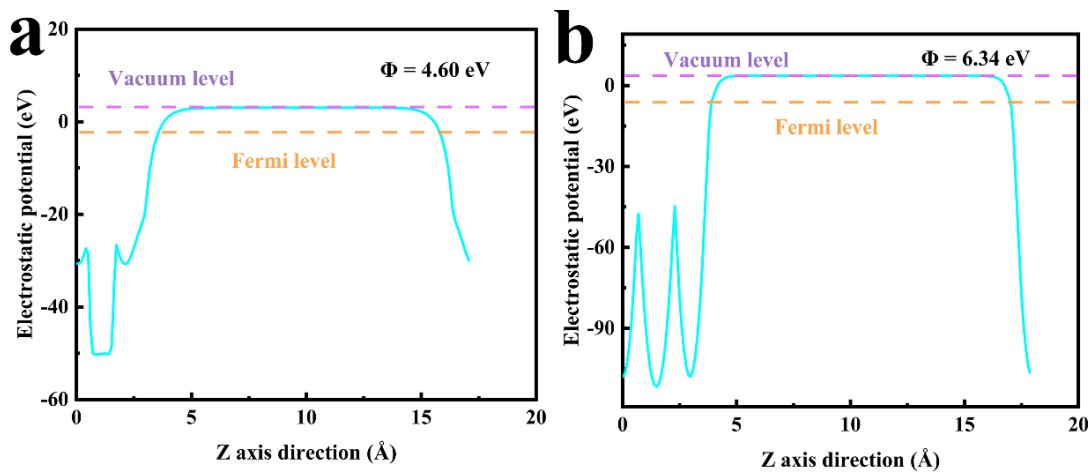


Figure S5. Electrostatic-potential distributions of (a) V₂CT_x and (b) SnS₂.

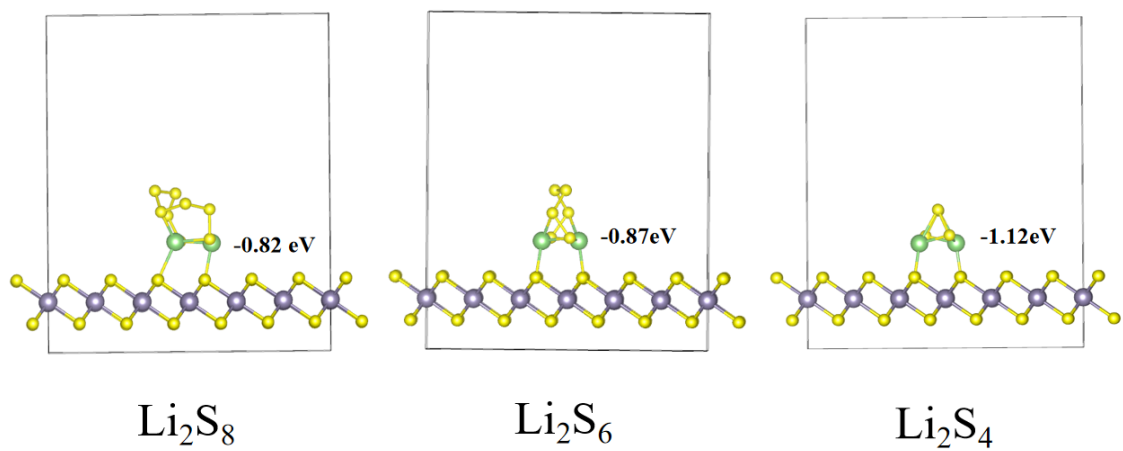


Figure S6. Calculated adsorption energies of Li₂S₈, Li₂S₆ and Li₂S₄ on SnS₂.

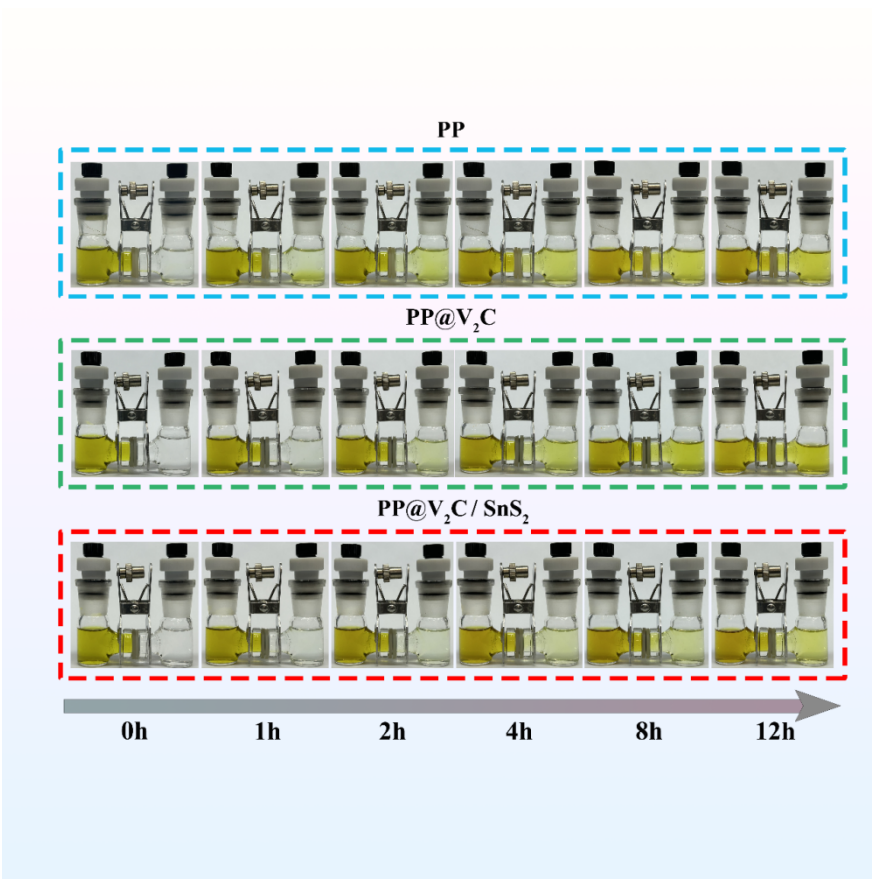


Figure S7. Photographs of Li₂S₆ diffusion through PP, PP@V₂CT_x and PP@V₂CT_x/SnS₂ separators at different times in an H-type cell.

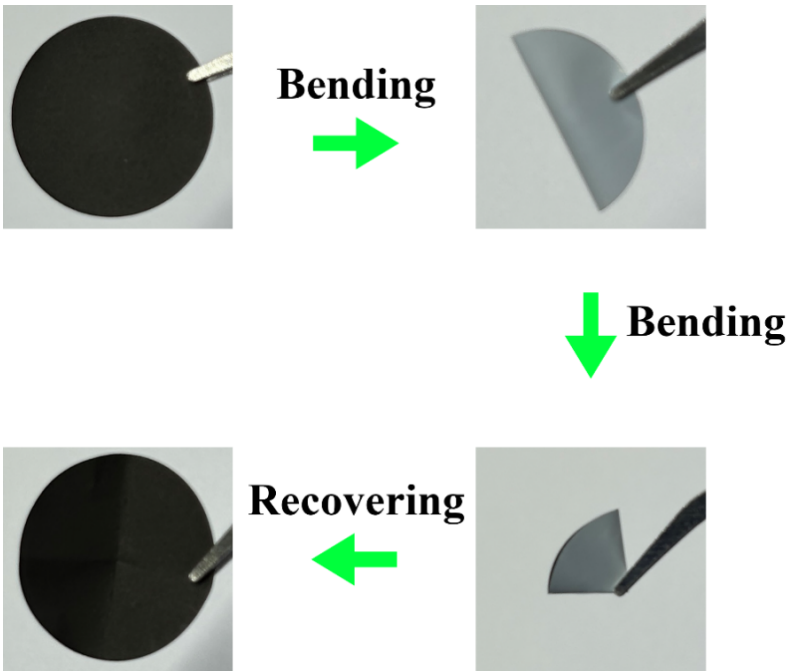


Figure S8. Photographs of the PP@V₂CT_x/SnS₂ separator in the bent state and after recovery.

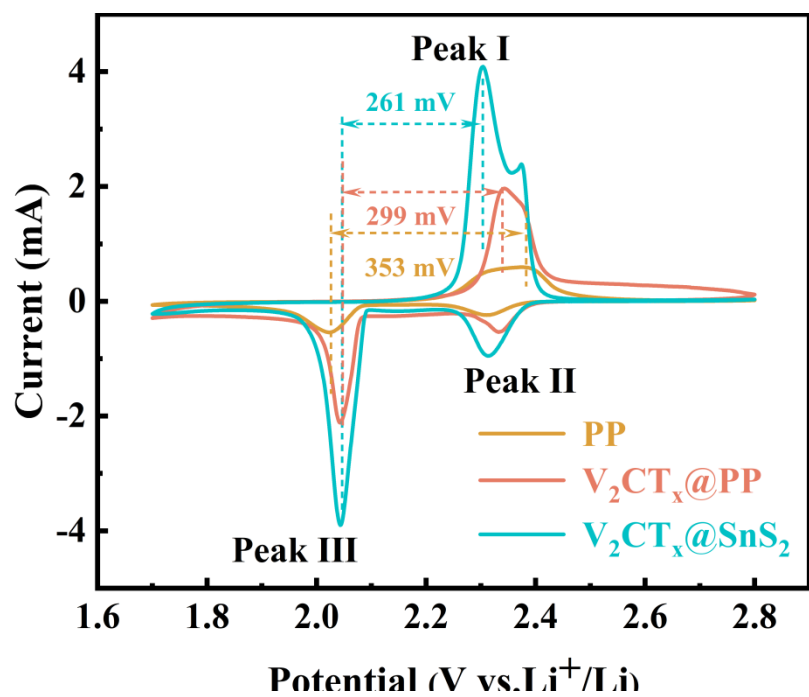


Figure S9. CV curves of cells with different separators at 0.1 mV s⁻¹.

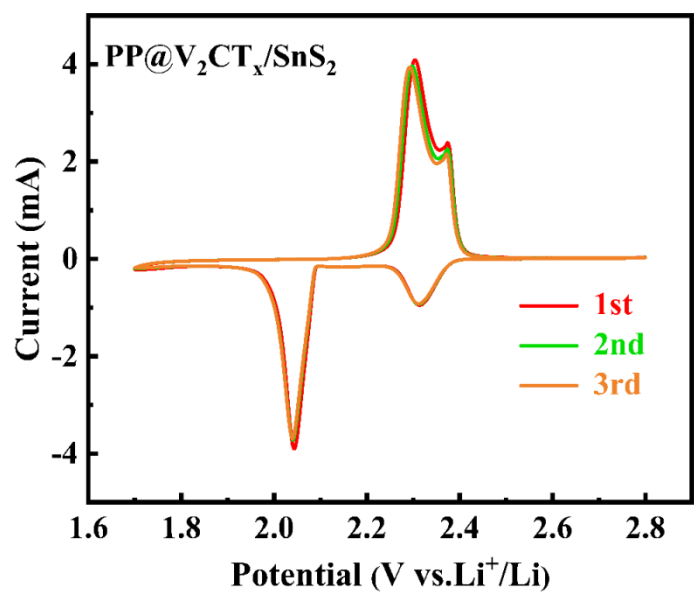


Figure S10. CV curves for the first three cycles of the $PP@V_2CT_x/SnS_2$ cell at 0.1 mV s⁻¹.

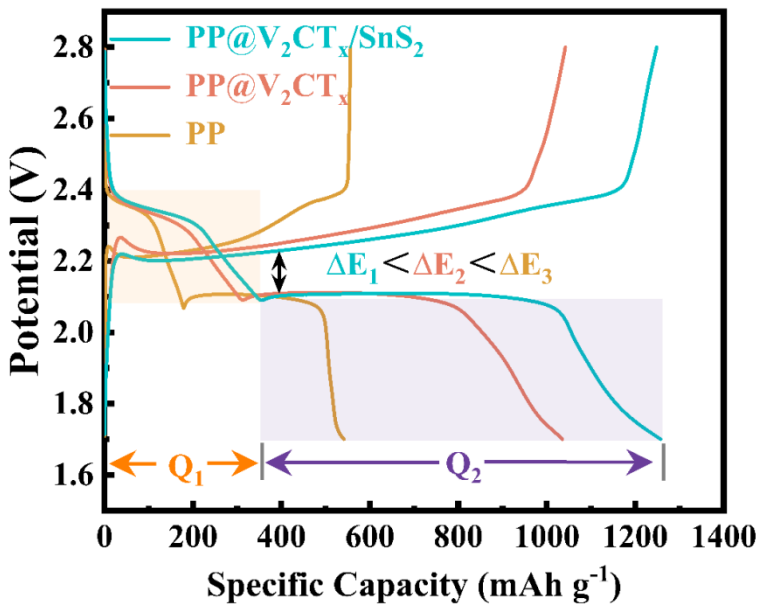


Figure S11. Galvanostatic charge–discharge profiles at 0.2 C.

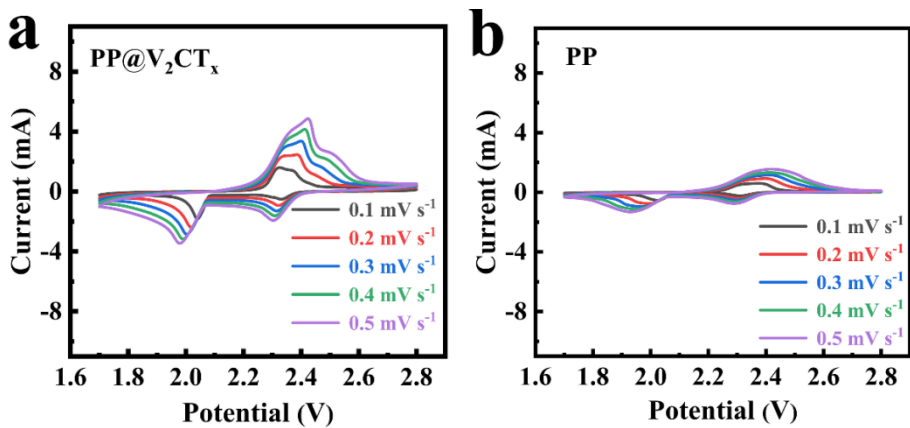


Figure S12. CV curves recorded from 0.1 to 0.5 mV s⁻¹ for cells with (a) PP@V₂CT_x and (b) PP separators.

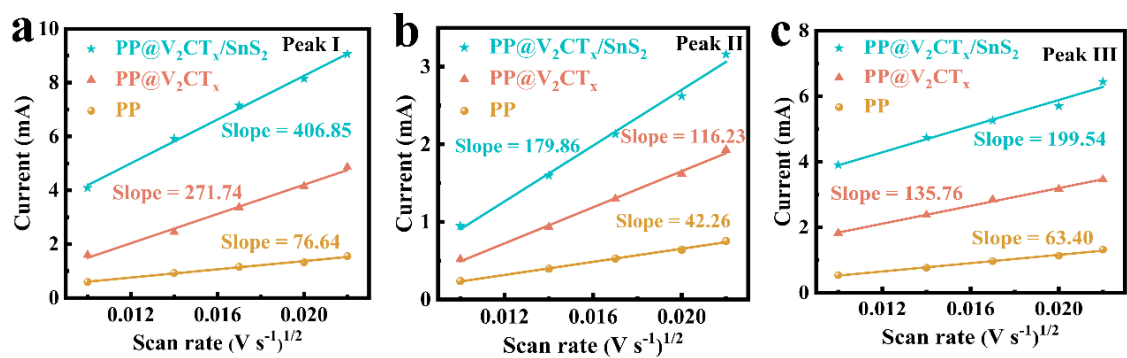


Figure S13. Linear fits of redox peak current versus the square root of scan rate for Peaks I–III, derived from Figure 3b and Figure S12a,b.

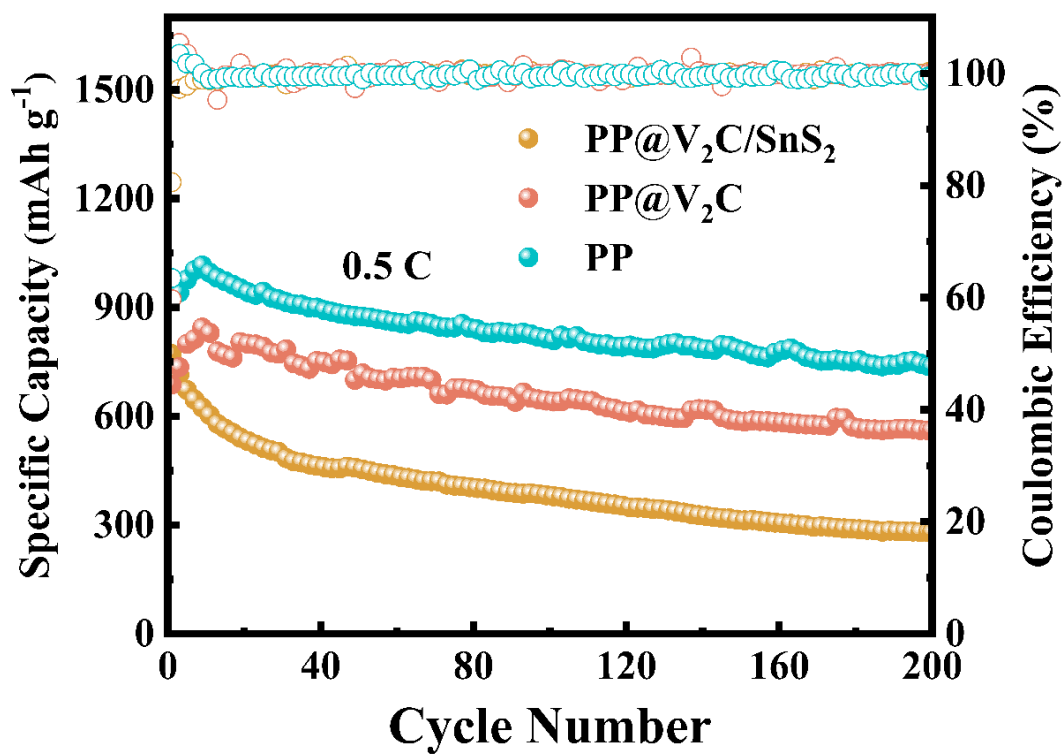


Figure S14. Cycling performance of Li–S cells with different separators at 0.5 C.

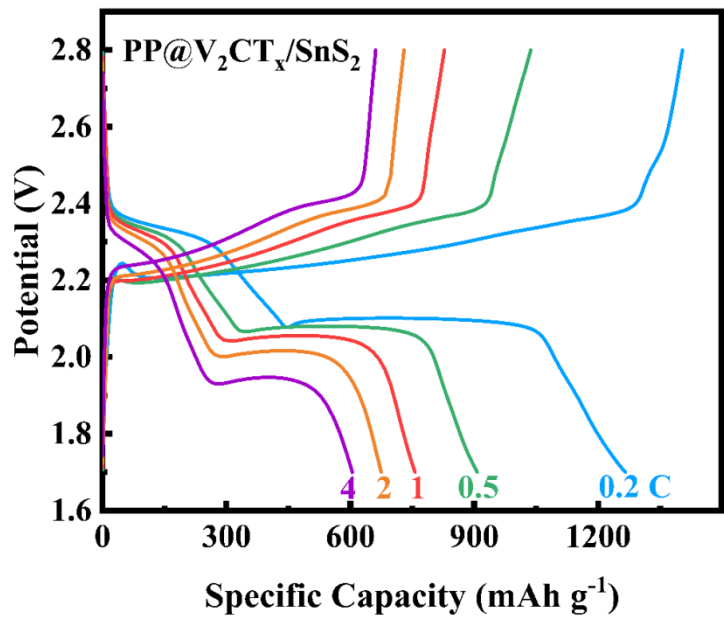


Figure S15. Galvanostatic charge–discharge profiles of the V_2CT_x/SnS_2 -separator cell at different rates.

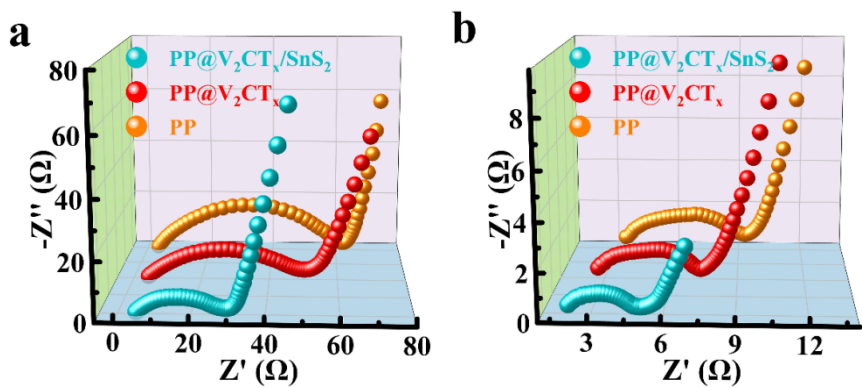


Figure S16. Nyquist plots of cells with different separators (a) before cycling and (b) after 1000 cycles at 1 C.

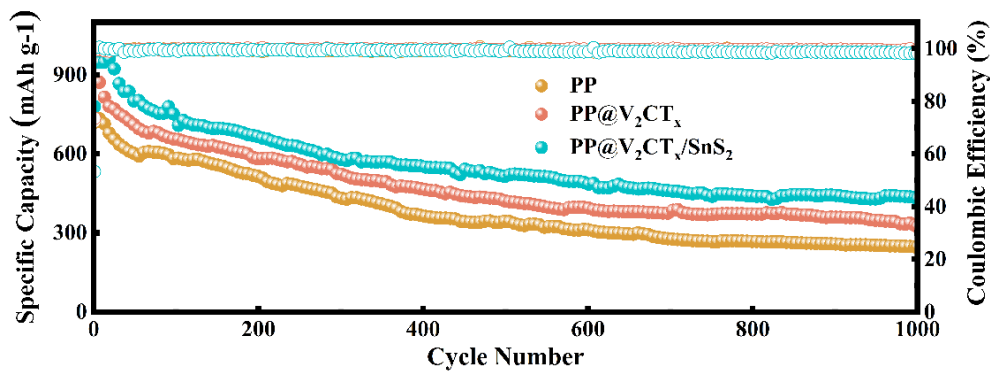


Figure S17. Long-term cycling performance of Li-S cells with different separators at 1 C.

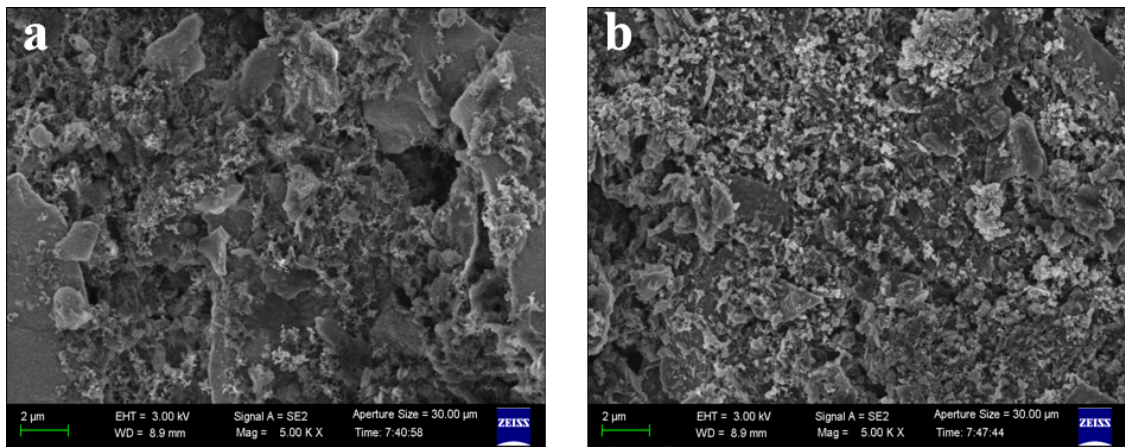


Figure S18. Top-view SEM images of the V₂CT_x/SnS₂/PP membrane (a) before and (b) after cycling.

S3. Supporting Table

Table S1. Comparison of the electrochemical performance of the present separator with previously reported systems.

Separator material	Sulfur loading (mg cm ⁻²)	Rate / capacity (C / mAh g ⁻¹)	Decay rate (% per cycle)
Ti ₃ C ₂ T _x /CNTs (10%) ^[4]	1.2	2 C/640	0.086
TOS/MX/TOS ^[5]	–	1 C/804.5	0.048
FeSA-PN@PNC ^[6]	1.5	2 C/659	0.040
MoS ₂ –x–Co ₉ S ₈ –y/rGO ^[7]	0.8–0.9	3 C/710.2	0.060
N-MX-CoS ₂ ^[8]	1.2	4 C/775	0.052
Ti ₃ C ₂ T _x /Nafion ^[9]	2	3 C/794	0.03
PM(0.4M)–CNT ^[10]	0.91	2 C/677.6	0.07
Ta ₄ C ₃ –Ta ₂ O ₅ @C ^[11]	1.2	1 C/738.5	0.086
OMC-g-MXene ^[12]	1.2	5 C/537	0.047
Co–MoS ₂ @NC ^[13]	1	4 C/610.1	0.053
STO-W/S ^[14]	0.8–1	2 C/492	0.067
MPF13-550/PP ^[15]	1.2–1.6	2 C/593	0.375
CoSe ₂ @TiSe ₂ –C ^[16]	2.5	3 C/717	0.031
7ZnO–3ZnS/rGO ^[17]	1	3 C/748.9	0.06
RG@CoS@C ^[18]	3	2 C/677	0.08
TiO ₂ @MoS ₂ ^[19]	1.0–1.2	5 C/601	0.067
VC-1 ^[20]	2.4–3	2 C/733.3	0.041
CC@VS ₂ –VO ₂ @Li ₂ S@C ^[21]	1.24	3 C/462.8	0.120
This work	1.08	4 C/653.0	0.019

S4. References

1. G. Kresse, D. Joubert, "From Ultrasoft Pseudopotentials to the Projector Augmented-Wave Method," *Physical Review B* 59, no. 3 (1999): 1758–1775. <https://doi.org/10.1103/PhysRevB.59.1758>
2. J. P. Perdew, K. Burke, M. Ernzerhof, "Generalized Gradient Approximation Made Simple," *Physical Review Letters* 77, no. 18 (1996): 3865–3868. <https://doi.org/10.1103/PhysRevLett.77.3865>
3. S. Grimme, J. Antony, S. Ehrlich, H. Krieg, "A Consistent and Accurate Ab Initio Parametrization of Density Functional Dispersion Correction (DFT-D) for the 94 Elements H–Pu," *The Journal of Chemical Physics* 132, no. 15 (2010): 154104. <https://doi.org/10.1063/1.3382344>
4. N. Li, Y. Xie, S. Peng, X. Xiong, K. Han, "Ultra-Lightweight $\text{Ti}_3\text{C}_2\text{T}_x$ MXene Modified Separator for Li–S Batteries: Thickness Regulation Enabled Polysulfide Inhibition and Lithium-Ion Transportation," *Journal of Energy Chemistry* 42 (2020): 116–125. <https://doi.org/10.1016/j.ijechem.2019.06.014>
5. Y. Yao, S. Wang, X. Jia, et al., "Freestanding Sandwich-Like Hierarchically TiS_2 – TiO_2 /MXene Bi-Functional Interlayer for Stable Li–S Batteries," *Carbon* 188 (2022): 533–542. <https://doi.org/10.1016/j.carbon.2021.12.036>
6. T. Huang, Y. Sun, J. Wu, et al., "Altering Local Chemistry of Single-Atom Coordination Boosts Bidirectional Polysulfide Conversion of Li–S Batteries," *Advanced Functional Materials* 32, no. 39 (2022): 2203902. <https://doi.org/10.1002/adfm.202203902>
7. H. Song, T. Li, T. He, et al., "Cooperative Catalytic Mo–S–Co Heterojunctions with Sulfur Vacancies for Kinetically Boosted Lithium–Sulfur Battery," *Chemical Engineering Journal* 450 (2022): 138115. <https://doi.org/10.1016/j.cej.2022.138115>
8. C. Yang, Y. Li, W. Peng, F. Zhang, X. Fan, "In Situ N-Doped CoS_2 Anchored on MXene toward an Efficient Bifunctional Catalyst for Enhanced Lithium–Sulfur Batteries," *Chemical Engineering Journal* 427 (2022): 131792. <https://doi.org/10.1016/j.cej.2021.131792>
9. J. Wang, P. Zhai, T. Zhao, et al., "Laminar MXene–Nafion-Modified Separator with Highly Inhibited Shuttle Effect for Long-Life Lithium–Sulfur Batteries," *Electrochimica Acta* 320 (2019): 134558. <https://doi.org/10.1016/j.electacta.2019.134558>
10. D. Xiong, S. Huang, D. Fang, et al., "Porosity Engineering of MXene Membrane towards Polysulfide Inhibition and Fast Lithium-Ion Transportation for Lithium–Sulfur Batteries," *Small* 17, no. 34 (2021): 2007442. <https://doi.org/10.1002/sml.202007442>
11. Q. Liang, S. Wang, X. Jia, et al., "MXene-Derivative Ta_4C_3 – Ta_2O_5 Heterostructure as a Bi-Functional Barrier for Li–S Batteries," *Journal of Materials Science & Technology* 151 (2023): 89–98. <https://doi.org/10.1016/j.jmst.2022.11.059>
12. X. Li, Q. Guan, Z. Zhuang, et al., "Ordered Mesoporous Carbon-Grafted MXene Catalytic Heterostructure as a Li-Ion Kinetic Pump toward High-Efficiency Sulfur/Sulfide Conversions for Li–S Batteries," *ACS Nano* 17, no. 2 (2023): 1653–1662. <https://doi.org/10.1021/acsnano.2c11663>
13. Y. Gong, Y. Wang, Z. Fang, et al., "Constructing a Catalytic Reservoir Using Cobalt Nanoparticles– MoS_2 @Nitrogen-Doped Carbon Nanotubes on the Separator to Immobilize Polysulfides and Accelerate Their Conversion for Lithium–Sulfur Batteries," *Chemical Engineering Journal* 446 (2022): 136943. <https://doi.org/10.1016/j.cej.2022.136943>
14. Z.-D. Huang, M.-T. Yang, J.-Q. Qi, et al., "Mitigating the Polysulfide Shuttle with TiO_2 Nanowire/Nanosheet Hybrid-Modified Separators for Robust Lithium–Sulfur Batteries," *Chemical Engineering Journal* 387 (2020): 124080. <https://doi.org/10.1016/j.cej.2020.124080>
15. L.-W. Lin, M. Qi, Z.-T. Bai, et al., "Crumpled Nitrogen-Doped Aerogels Derived from MXene and Pyrrole–Formaldehyde as Modified Separators for Stable Lithium–Sulfur Batteries," *Applied Surface Science* 555 (2021): 149717. <https://doi.org/10.1016/j.apsusc.2021.149717>
16. L. Wang, X. Meng, X. Wang, M. Zhen, "Dual-Conductive CoSe_2 @ TiSe_2 –C Heterostructures Promoting Overall Sulfur Redox Kinetics under High Sulfur Loading and Lean Electrolyte," *Small* 19, no. 21 (2023): 2300089. <https://doi.org/10.1002/sml.202300089>
17. W. Zhang, D. Hong, Z. Su, et al., "Tailored ZnO – ZnS Heterostructure Enables a Rational Balance of Strong Adsorption and High Catalytic Activity of Polysulfides for Li–S Batteries," *Energy Storage Materials* 53 (2022): 404–414. <https://doi.org/10.1016/j.ensm.2022.09.018>
18. J. Guo, H. Jiang, X. Li, et al., "Defective Graphene-Coating-Induced Exposed Interfaces on CoS Nanosheets for High Redox Electrocatalysis in Lithium–Sulfur Batteries," *Energy Storage Materials* 40 (2021): 358–367. <https://doi.org/10.1016/j.ensm.2021.05.031>
19. J. Lee, C. Choi, J. B. Park, et al., "Optimally Arranged TiO_2 @ MoS_2 Heterostructures with an Effectively Induced Built-In Electric Field for High-Performance Lithium–Sulfur Batteries," *Journal of Energy Chemistry* 83 (2023): 496–508. <https://doi.org/10.1016/j.ijechem.2023.04.016>

WILEY-VCH

20. L. Chen, Y. Sun, X. Wei, et al., “Dual-Functional V₂C MXene Assembly in Facilitating Sulfur Evolution Kinetics and Li-Ion Sieving toward Practical Lithium–Sulfur Batteries,” *Advanced Materials* 35, no. 26 (2023): 2300771. <https://doi.org/10.1002/adma.202300771>
21. Y. Deng, W. Tang, Y. Zhu, et al., “Catalytic VS₂–VO₂ Heterostructure That Enables a Self-Supporting Li₂S Cathode for Superior Lithium–Sulfur Batteries,” *Small Methods* 7, no. 6 (2023): 2300186. <https://doi.org/10.1002/smt.202300186>